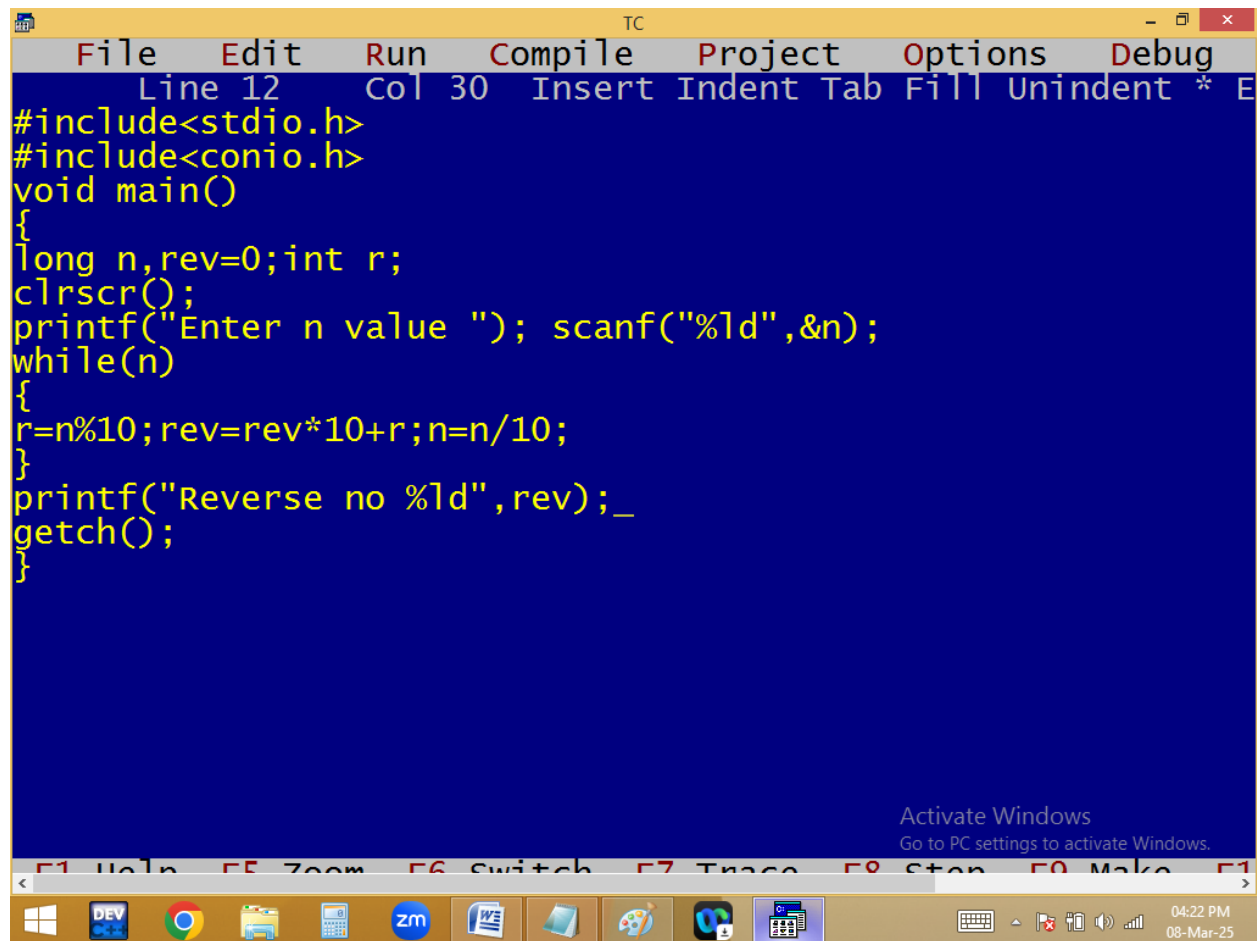


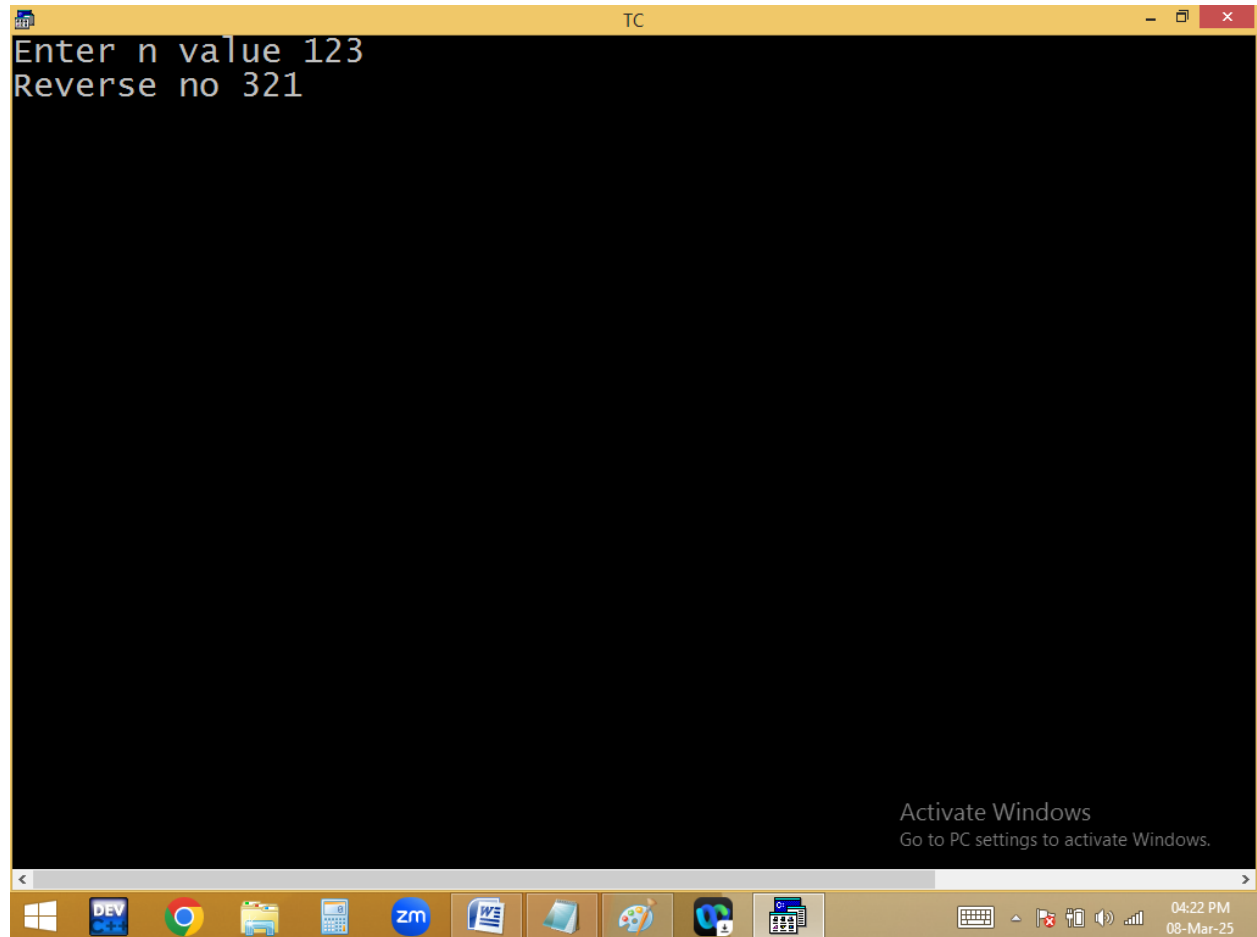
Reverse no:

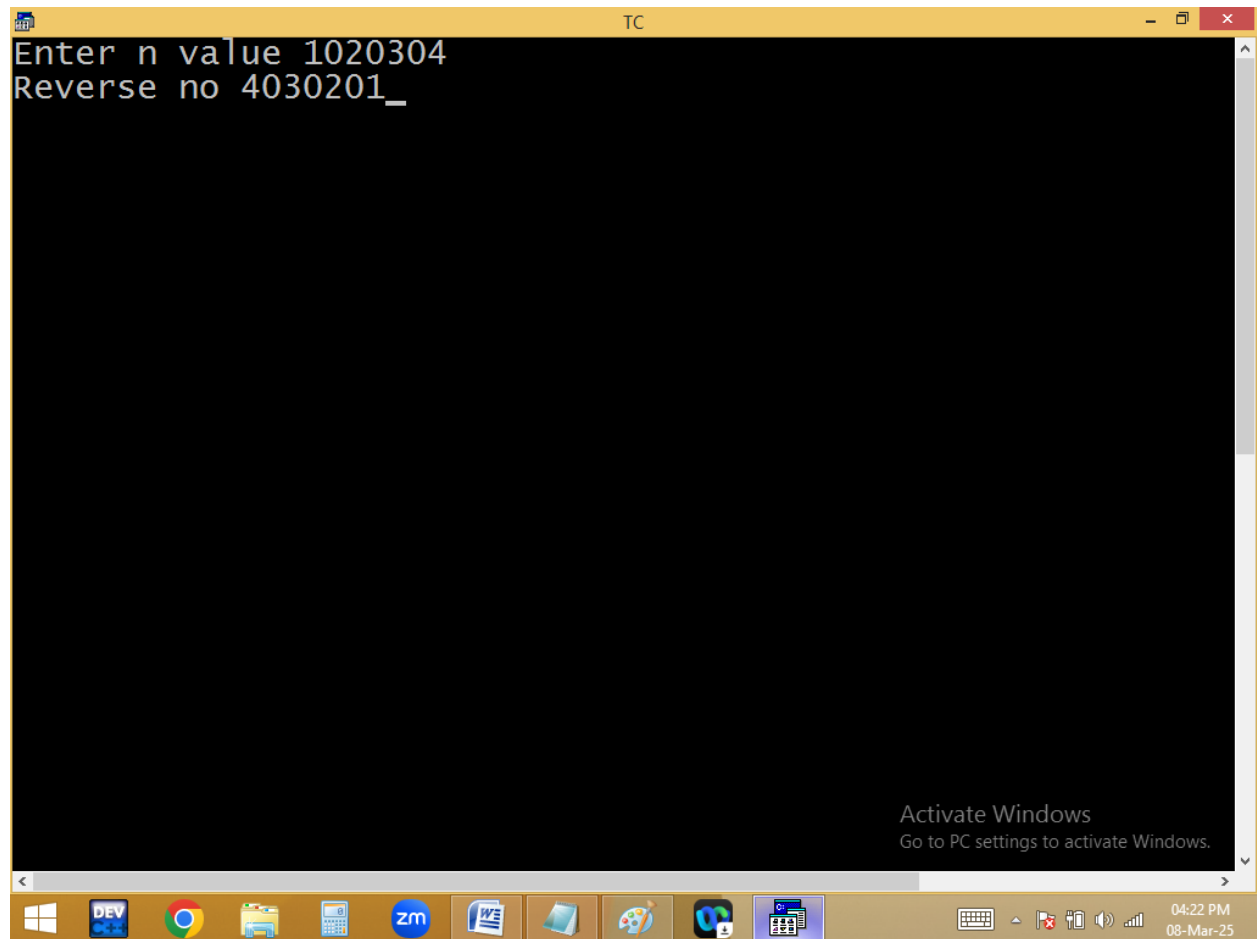


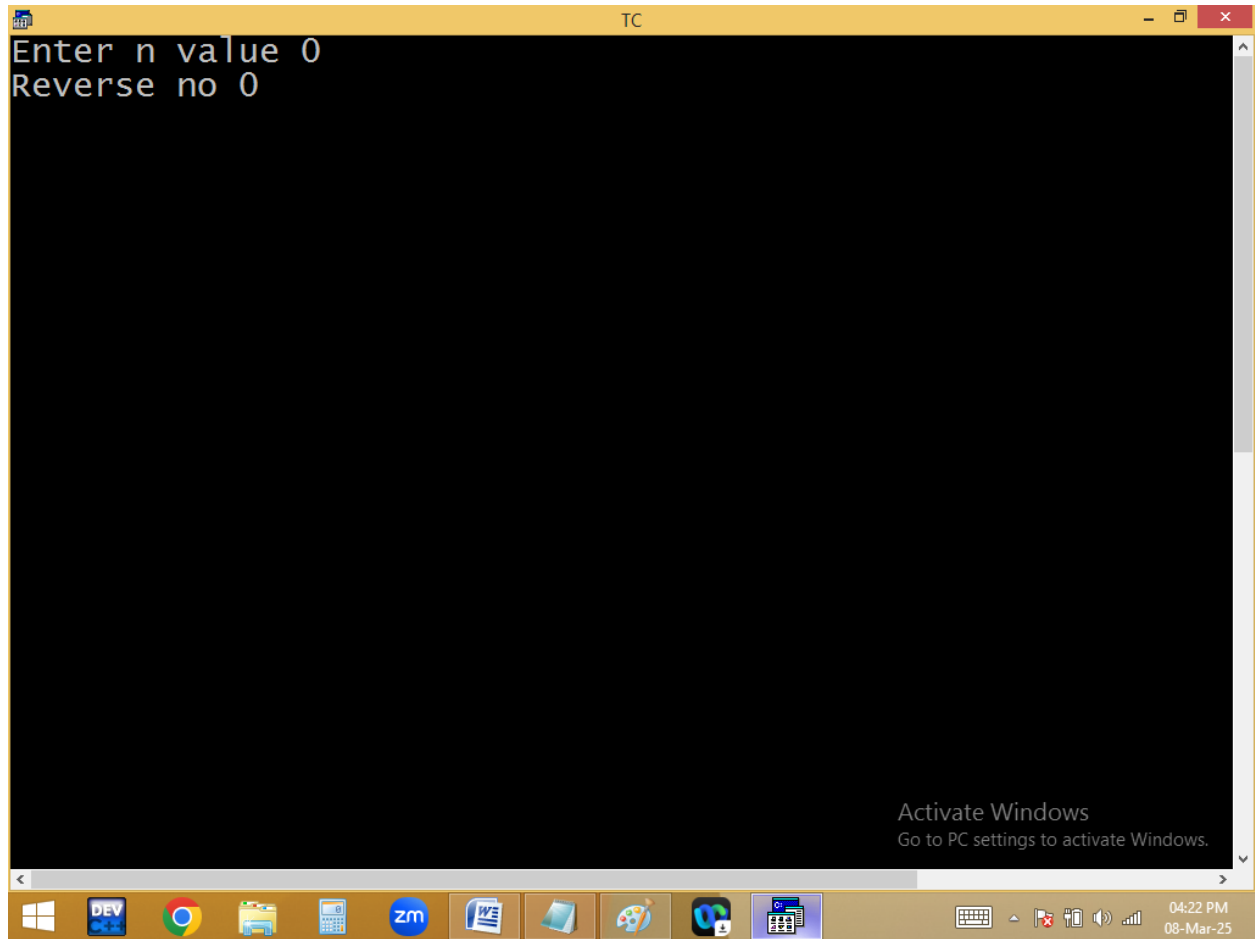
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status line indicates "Line 12 Col 30 Insert Indent Tab Fill Unindent * E". The main editing area has a dark blue background with yellow text. It contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    long n,rev=0;int r;
    clrscr();
    printf("Enter n value "); scanf("%ld",&n);
    while(n)
    {
        r=n%10;rev=rev*10+r;n=n/10;
    }
    printf("Reverse no %ld",rev);_
    getch();
}
```

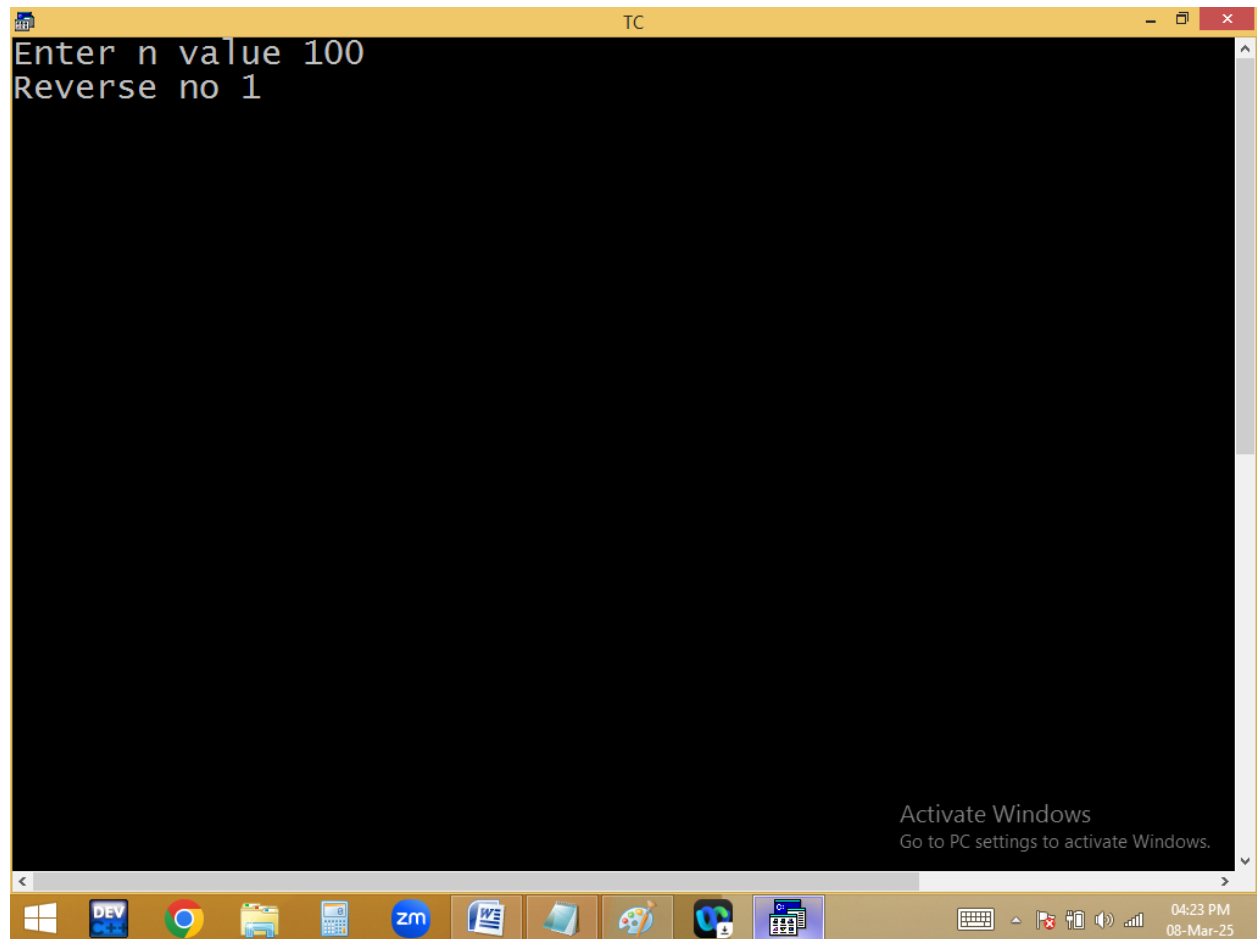
At the bottom of the window, there is a taskbar with icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, Word, and other applications. The system tray on the right shows the time as 04:22 PM and the date as 08-Mar-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.





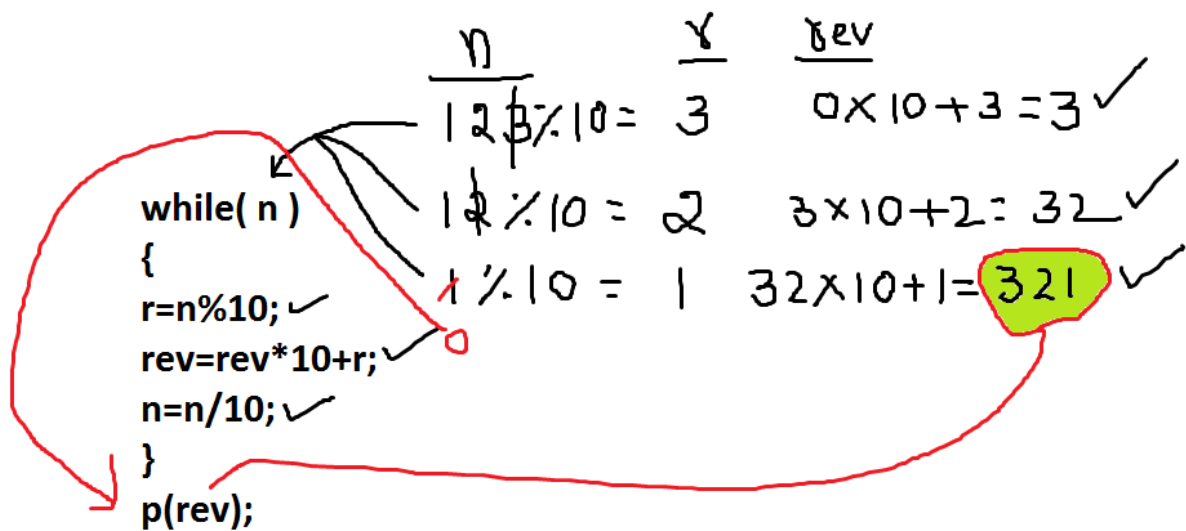






```
Enter n value 12000
Reverse no 21_
```

Activate Windows
Go to PC settings to activate Windows.



$$\frac{n}{} \quad \frac{r}{}$$

$$100 \% 10 = 0$$

$$10 \% 10 = 0$$

$$1 \% 10 = 1$$

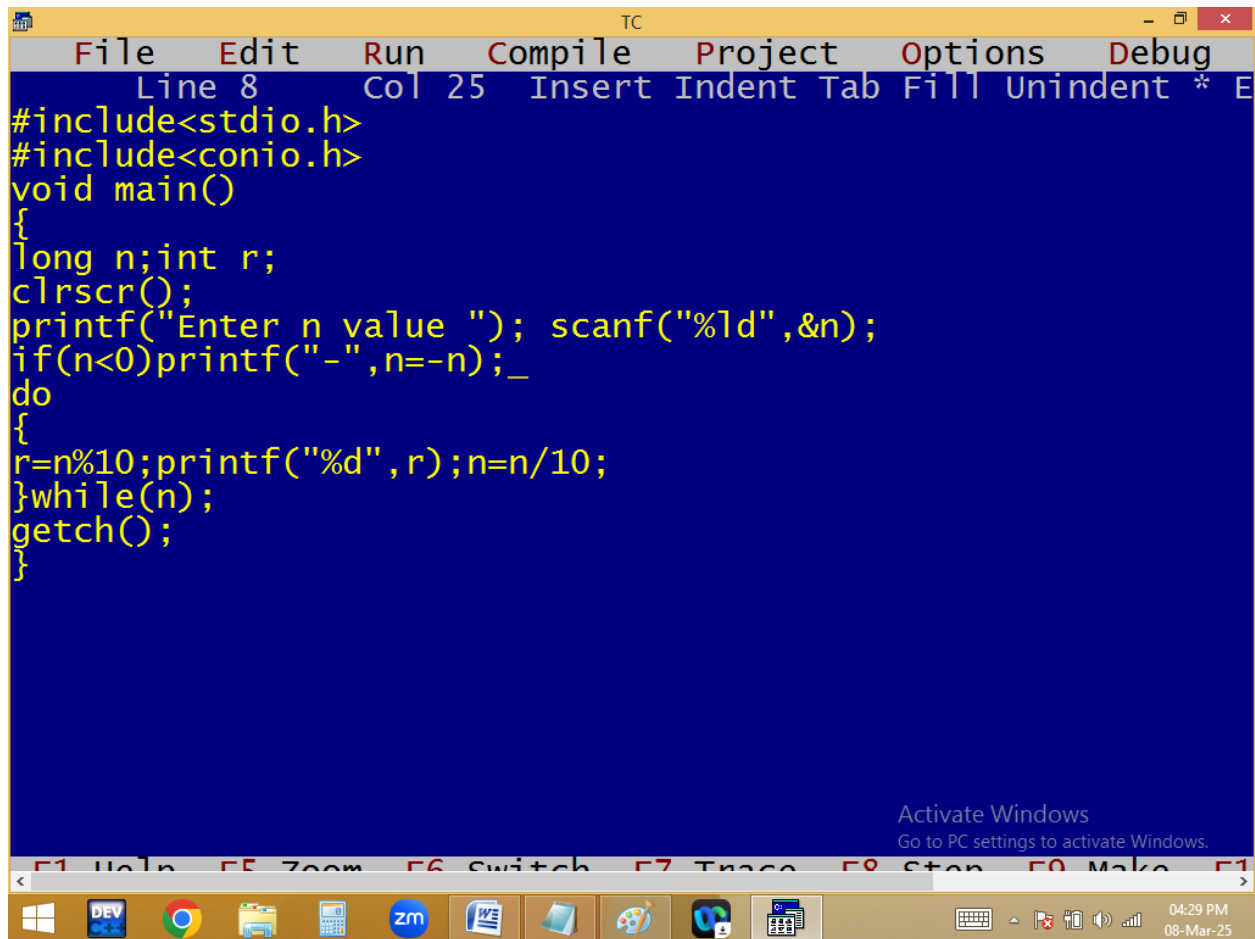
$$\frac{rev}{}$$

$$0 \times 10 + 0 = 0$$

$$0 \times 10 + 0 = 0$$

$$0 \times 10 + 1 = 1 \checkmark$$

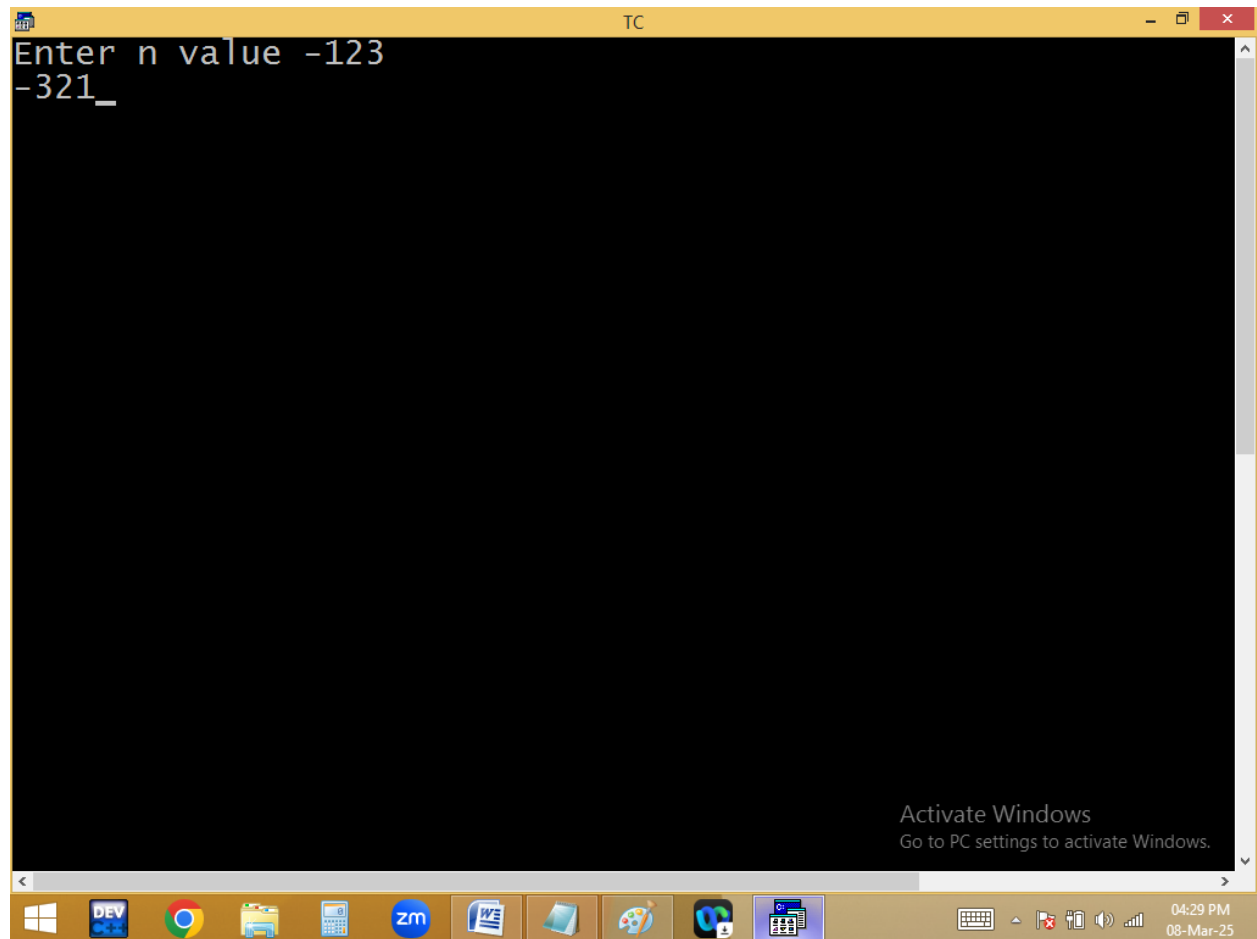
Printing 100 as 001:

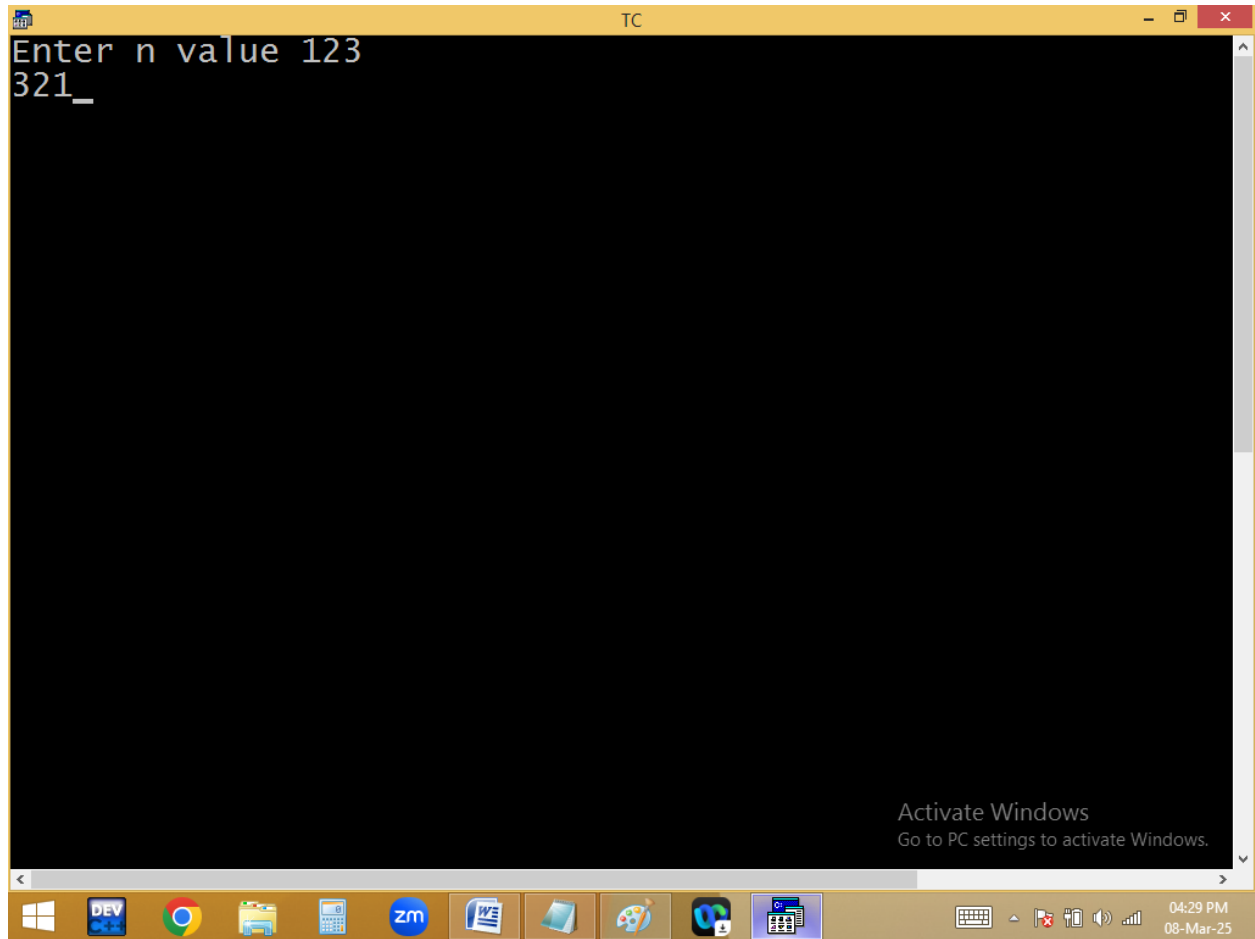


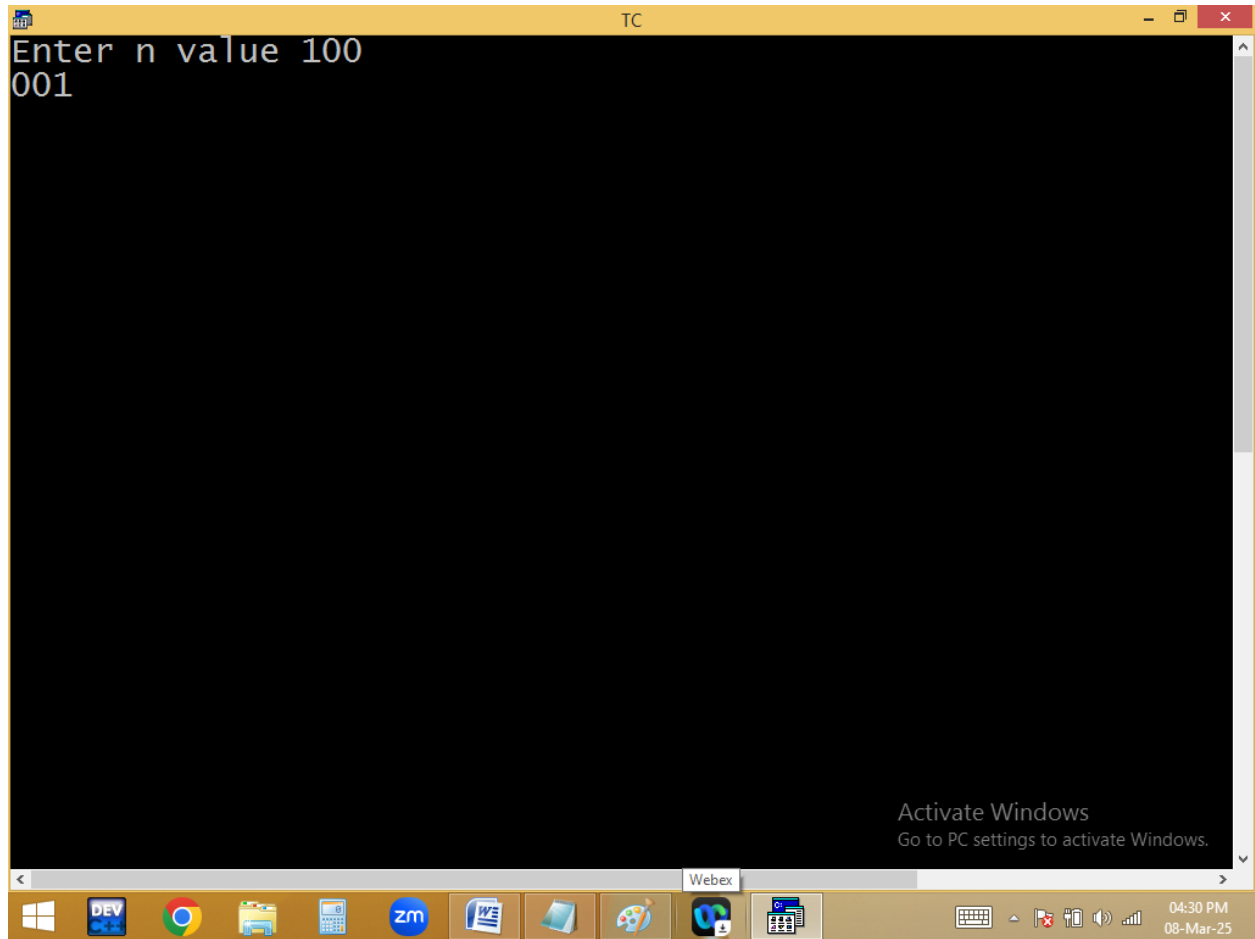
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 8", "Col 25", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background and contains the following C code:

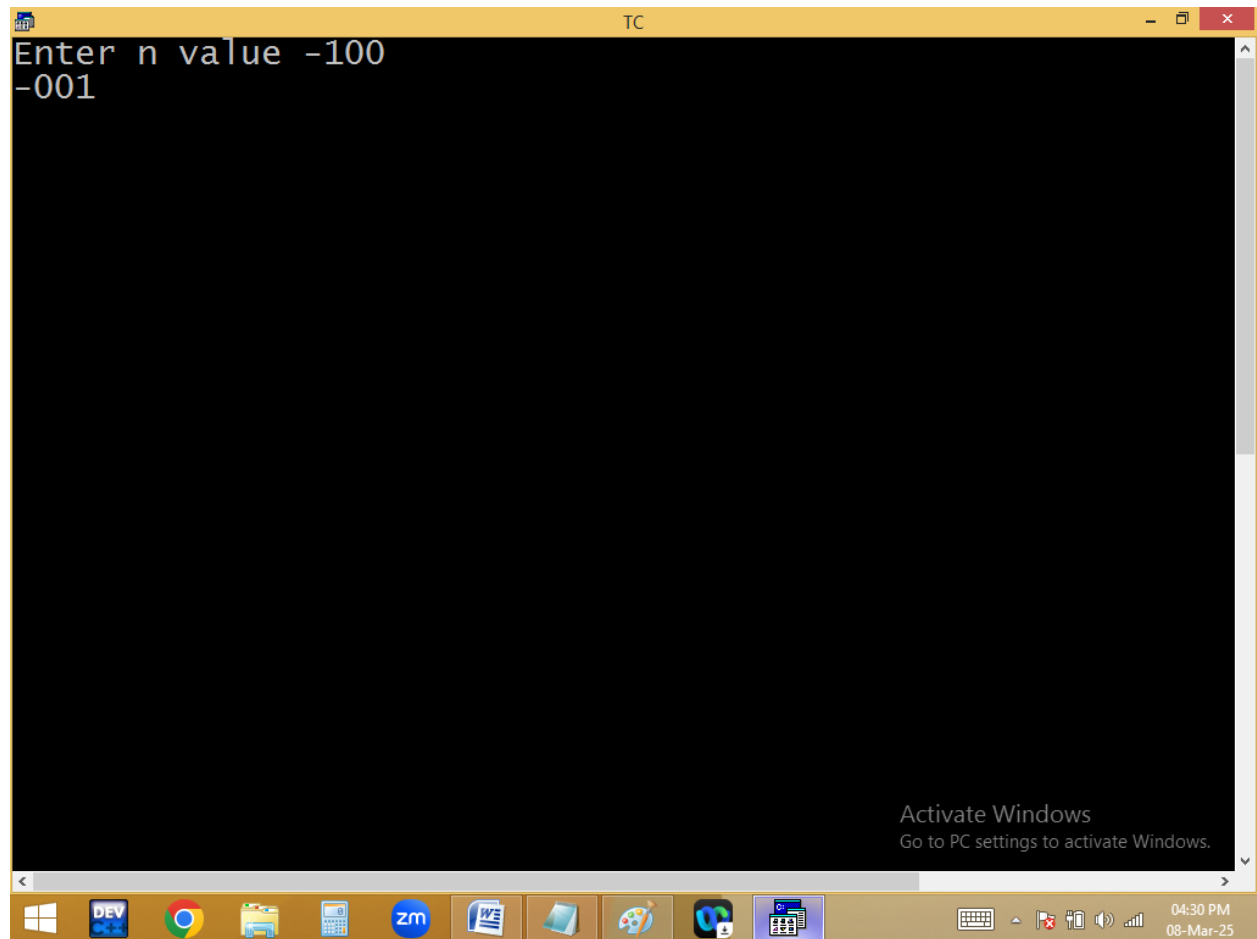
```
#include<stdio.h>
#include<conio.h>
void main()
{
    long n;int r;
    clrscr();
    printf("Enter n value "); scanf("%ld",&n);
    if(n<0)printf("-",n=-n);_
    do
    {
        r=n%10;printf("%d",r);n=n/10;
    }while(n);
    getch();
}
```

At the bottom of the IDE window, there is a text prompt: "Activate Windows Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which includes icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, Word, and other applications. The system tray on the right shows the time as "04:29 PM" and the date as "08-Mar-25".







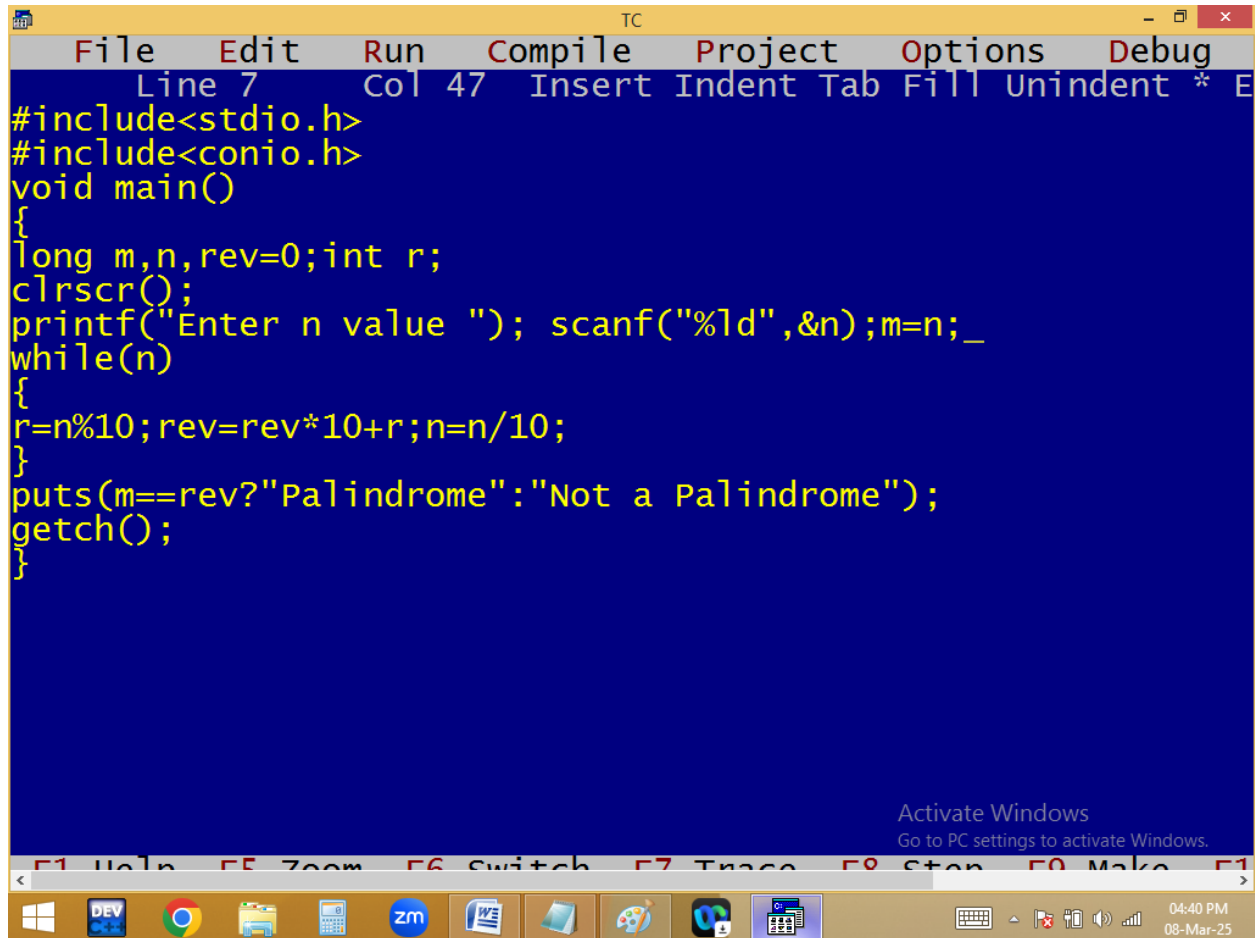


```
TC
Enter n value 0
0
```

Activate Windows
Go to PC settings to activate Windows.

$$\begin{array}{l} n \quad \text{print}(x) \\ \hline 100 \% 10 = 0 \\ 10 \% 10 = 0 \\ 1 \% 10 = 1 \end{array}$$

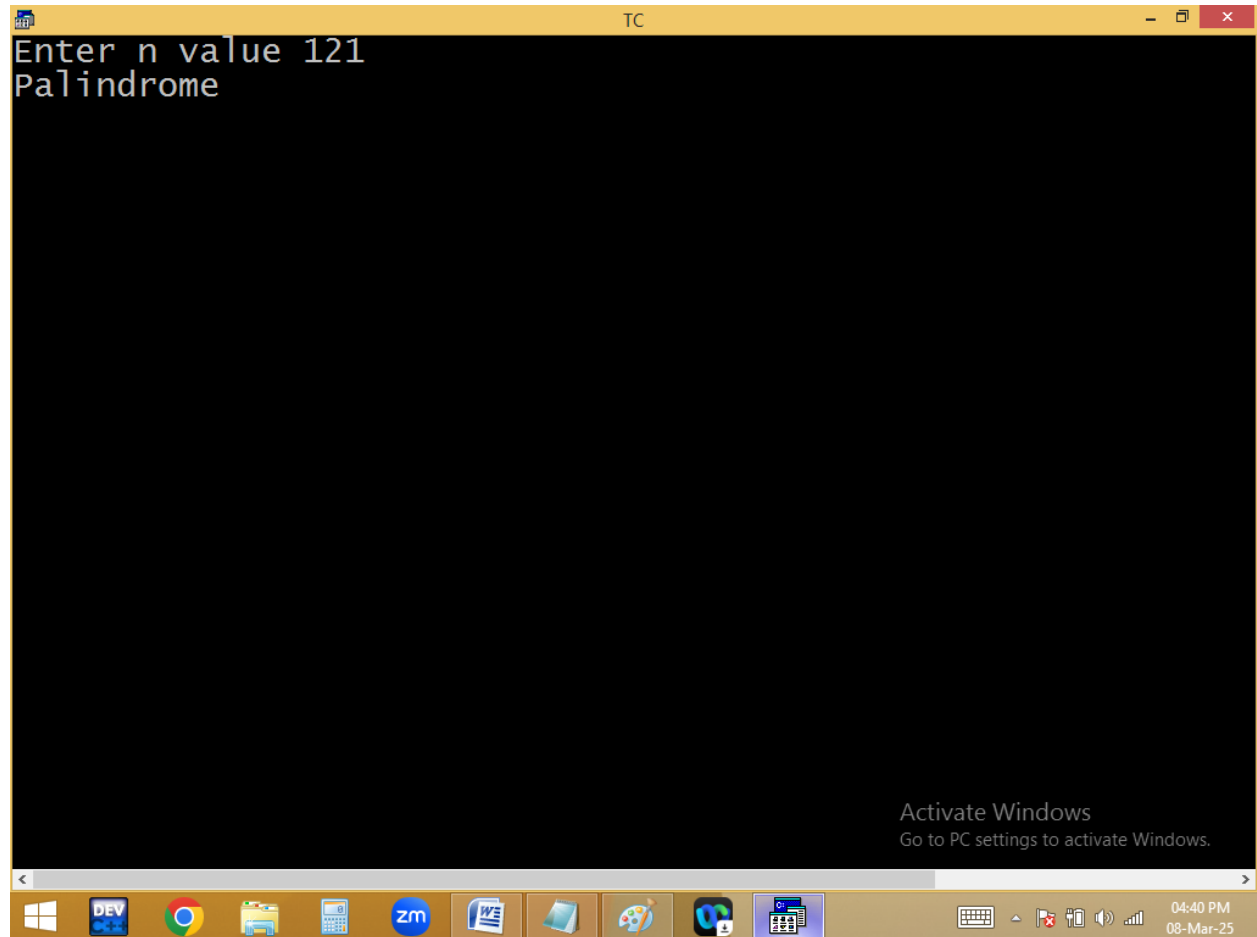
Finding palindrome or not?

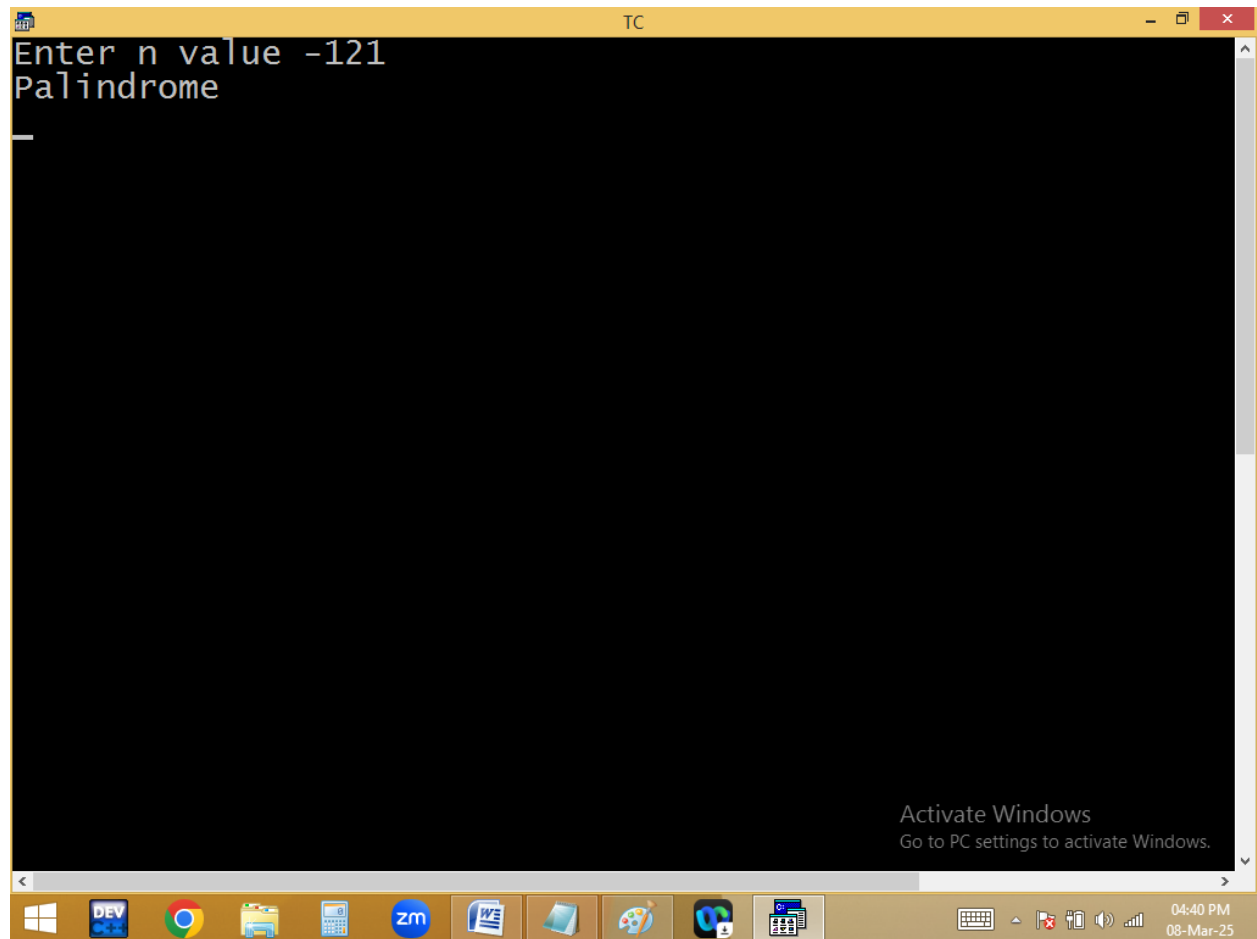


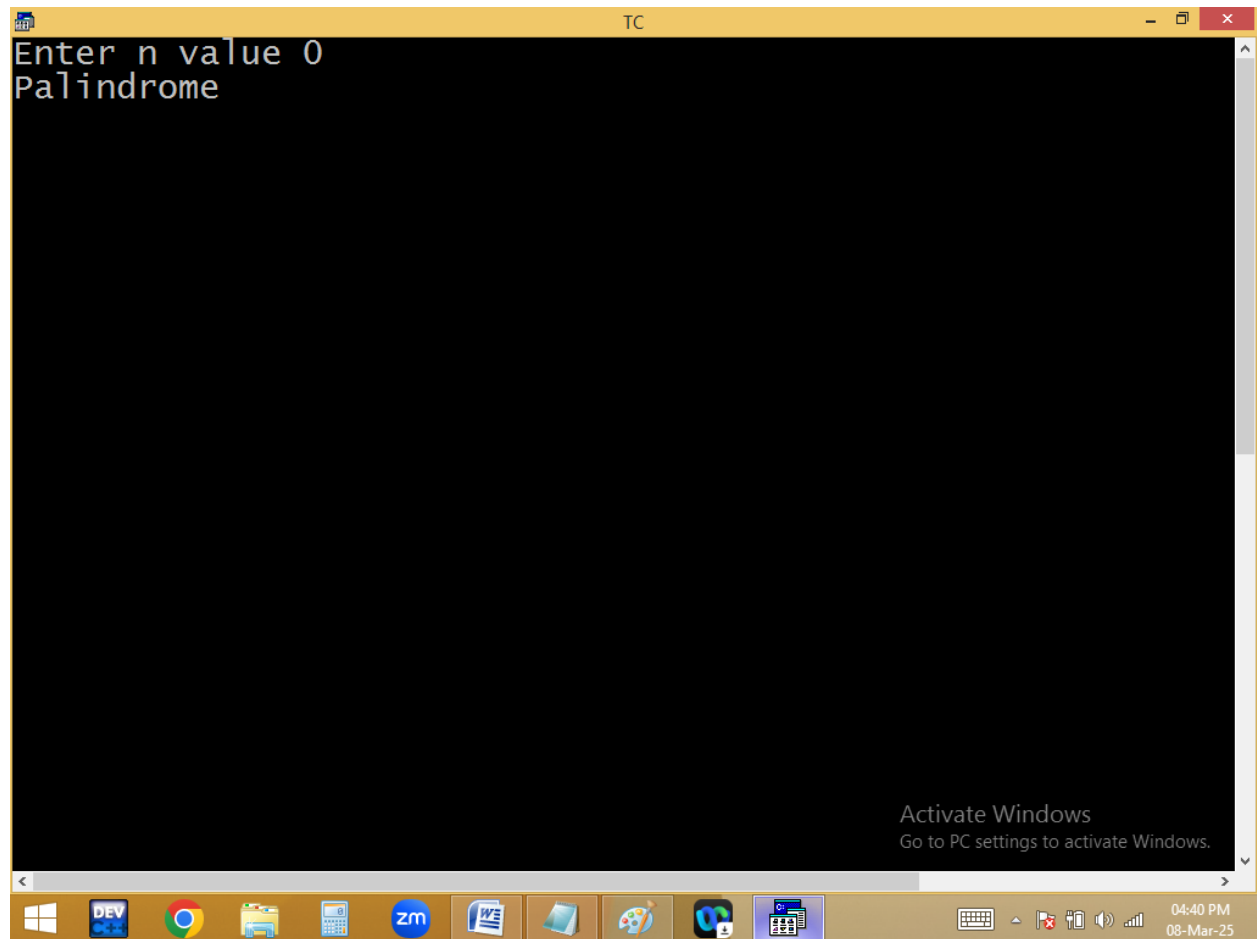
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 7", "Col 47", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", and a cursor icon. The main editing area has a dark blue background with yellow text. It contains a C program that checks if a number is a palindrome. The code is as follows:

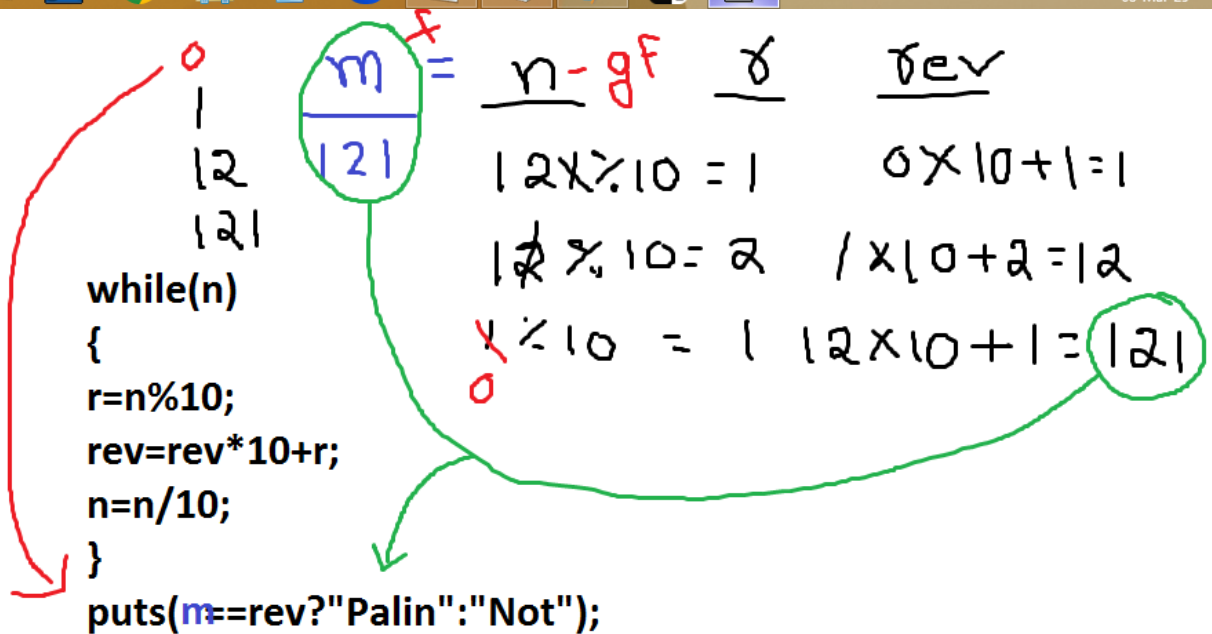
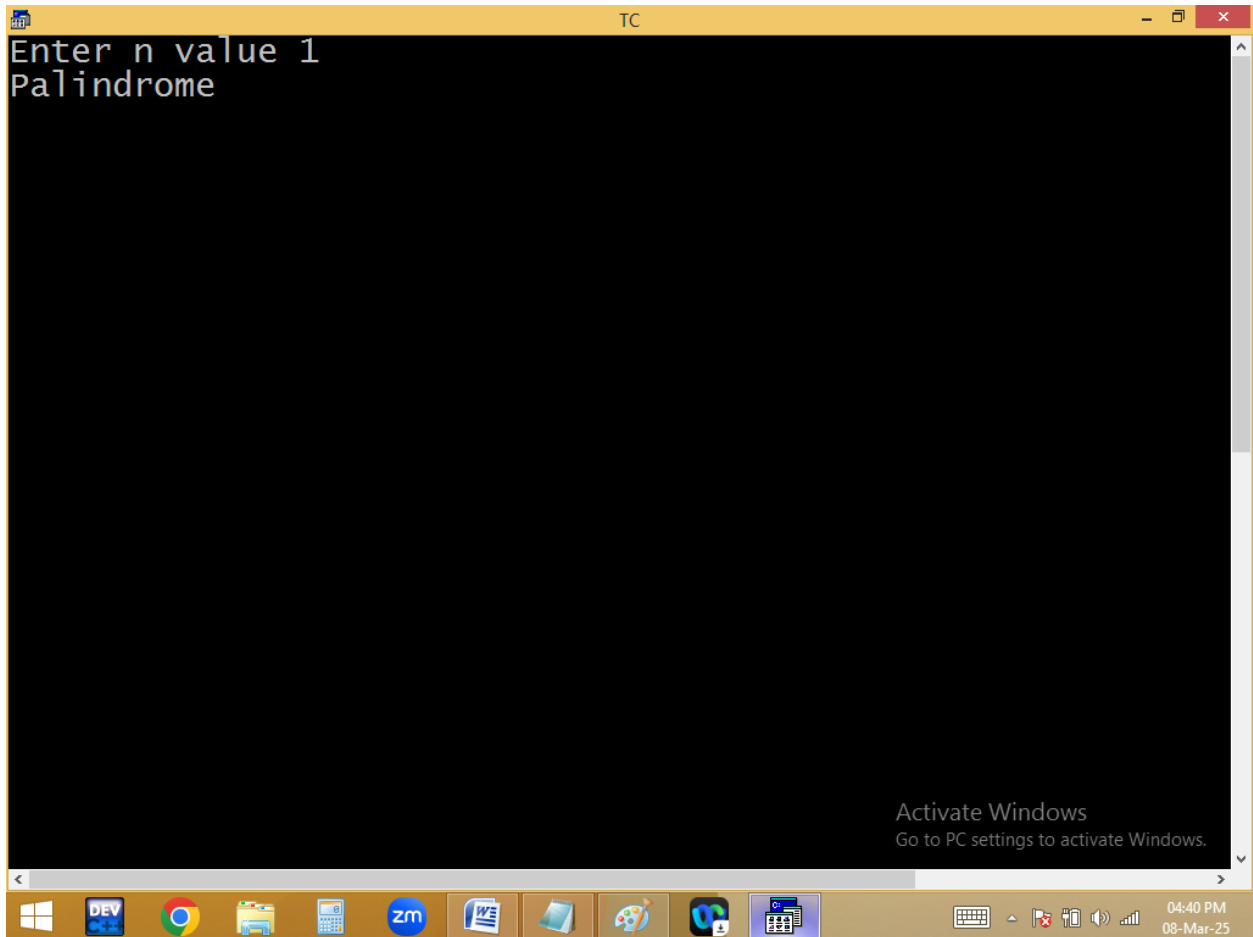
```
#include<stdio.h>
#include<conio.h>
void main()
{
    long m,n,rev=0;int r;
    clrscr();
    printf("Enter n value "); scanf("%ld",&n);m=n;_
    while(n)
    {
        r=n%10;rev=rev*10+r;n=n/10;
    }
    puts(m==rev?"Palindrome":"Not a Palindrome");
    getch();
}
```

At the bottom of the IDE window, there is a toolbar with icons for various functions, and a status bar showing the time "04:40 PM" and date "08-Mar-25". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.









Finding spy no or not?

$$123=1+2+3=6$$

$$1*2*3=6$$

$$1412=1+4+1+2=8$$

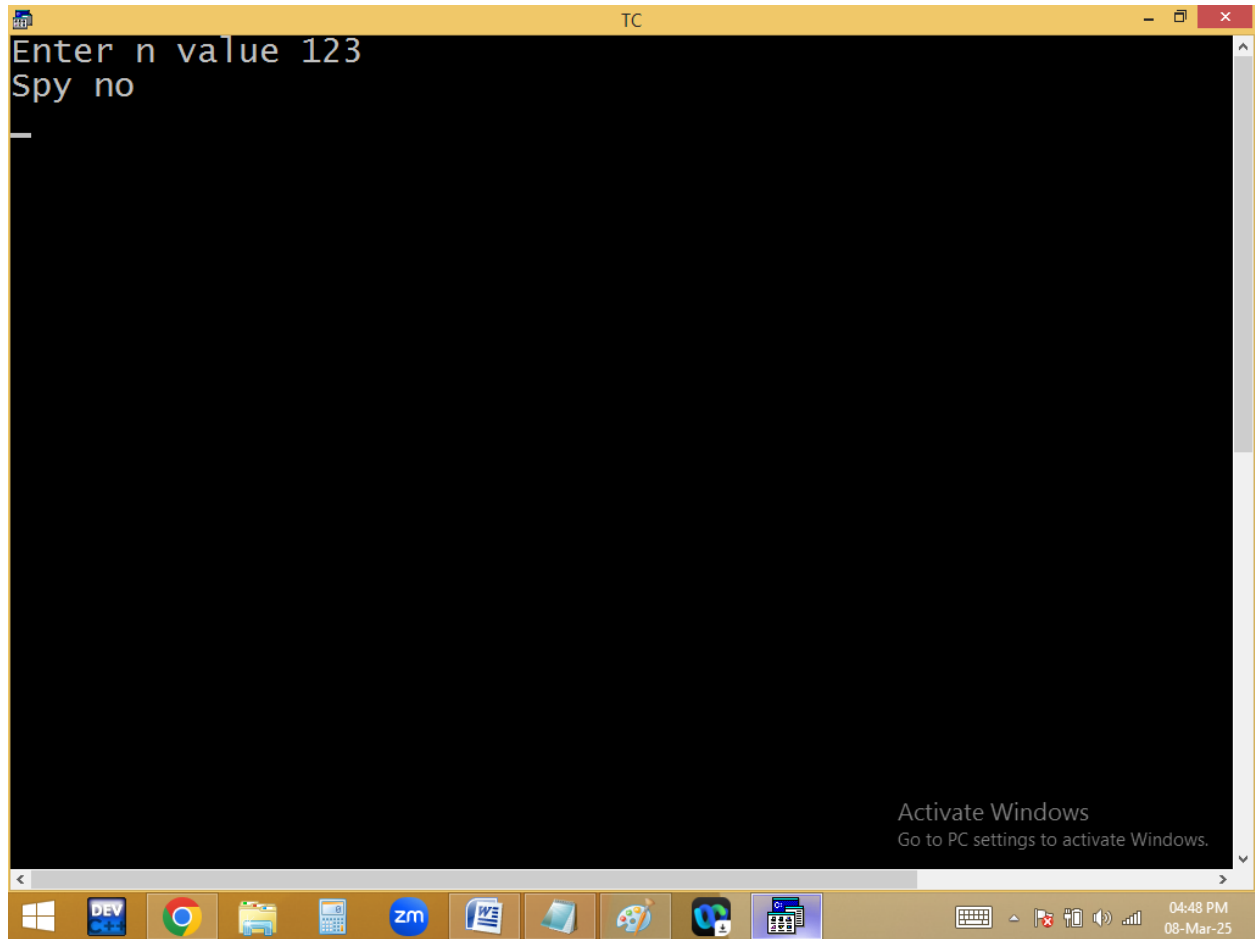
$$1*4*1*2=8$$

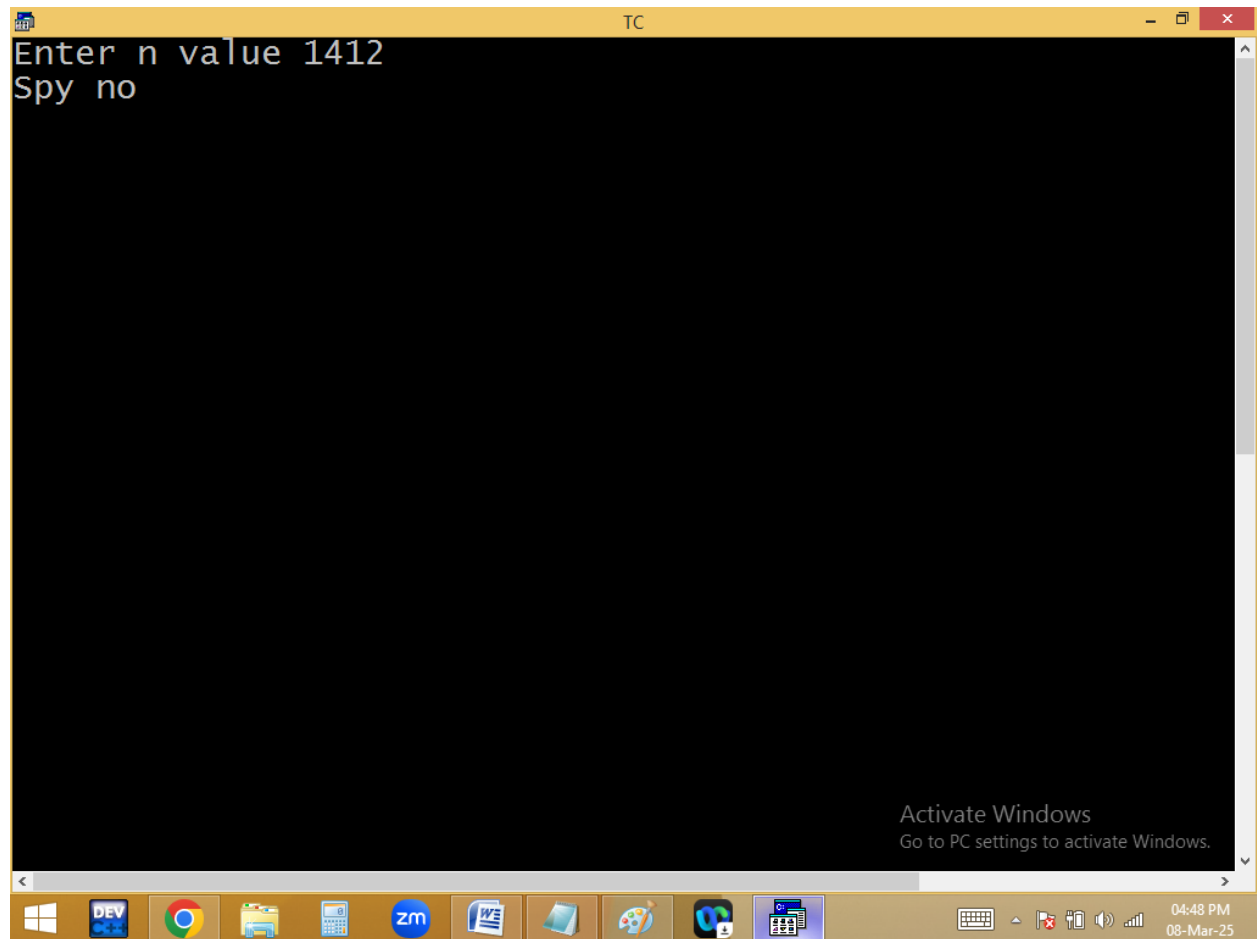
```
TC
File Edit Run Compile Project Options Debug
Line 7 Col 43 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
long n;int r,s1=0,s2=1;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
while(n)
{
r=n%10;s1+=r;s2*=r;n=n/10;
}
puts(s1==s2?"Spy no":"Not a Spy no");
getch();
}
```

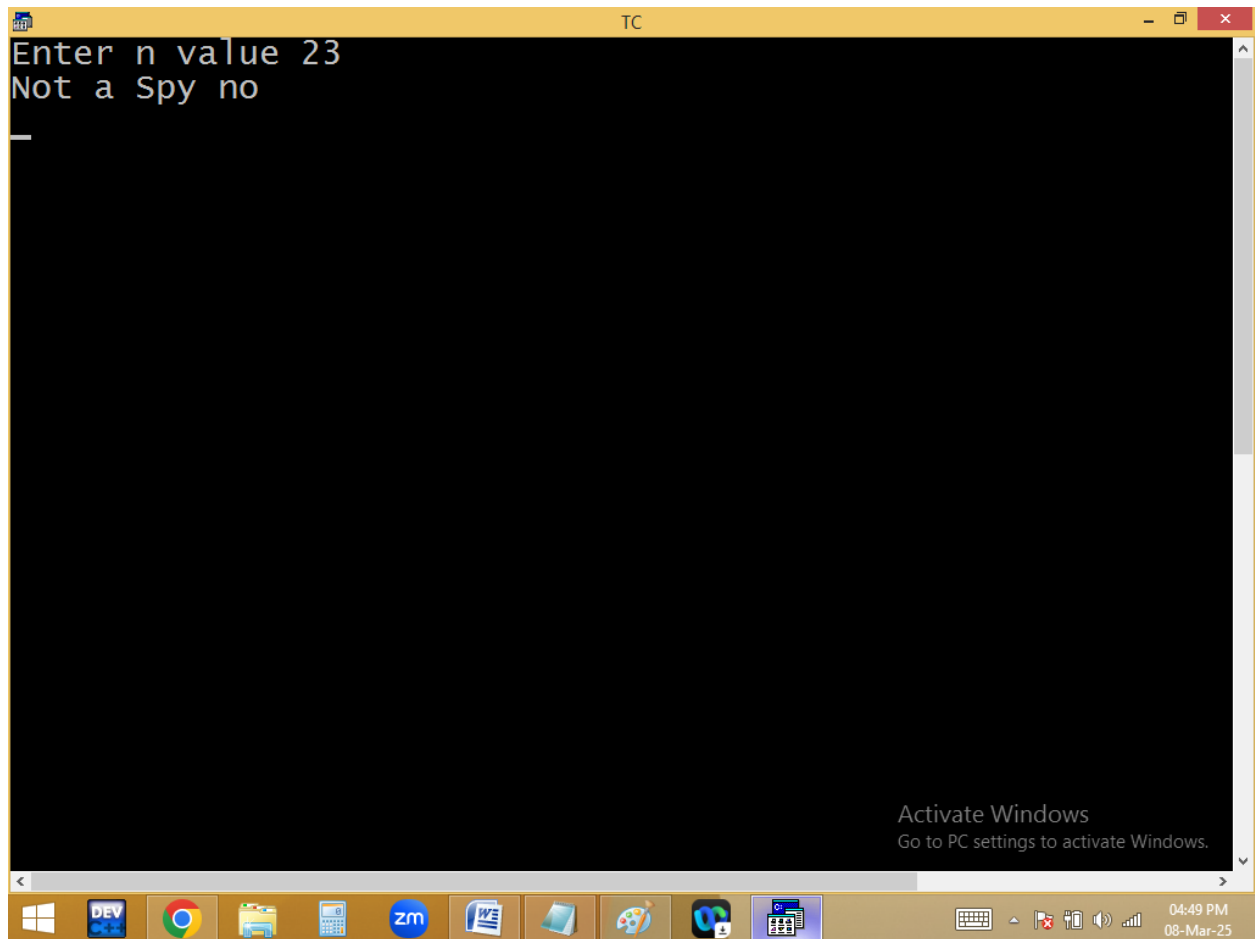
Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

04:48 PM
08-Mar-25







~~0~~
~~1412~~

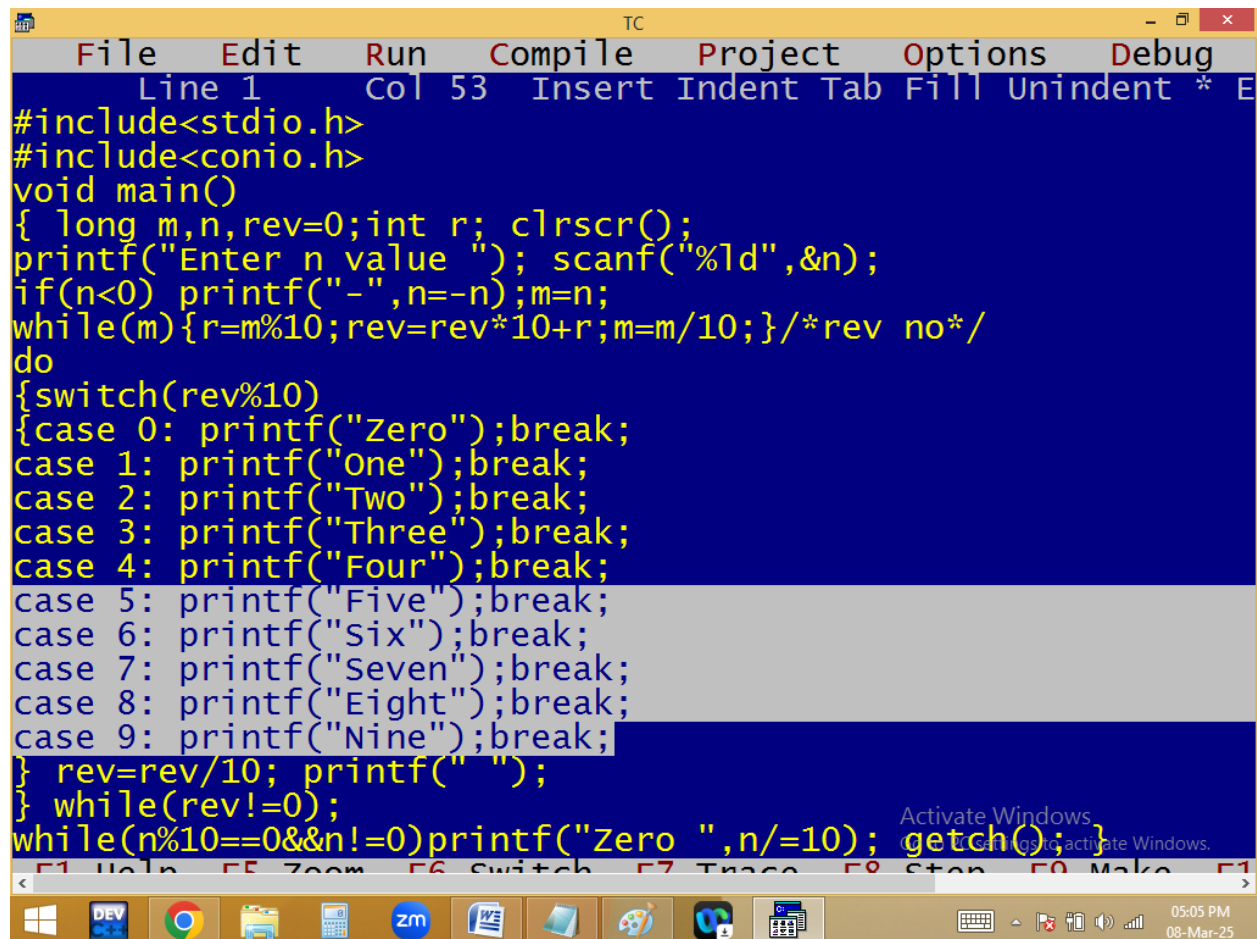
```
while( n )  
{  
  r = n %10;  
  s1+=r;  
  s2*=r;  
  n /=10;  
}  
puts(s1==s2?"Spy":"Not");
```

<u>n</u>	<u>s1</u>	<u>s2</u>
1412 % 10 = 2	0 + 2	1 × 2 × 1 × 4 × 1
141 % 10 = 1	+ 1	<u>8</u>
14 % 10 = 4	+ 4	
1 % 10 = 1	<u>+ 1</u>	
	8	

Red arrows indicate the flow of the algorithm: from the initial value of $s2$ (1) to the final result (8), and from the final result (8) to the output statement in the code.

No to text conversion:

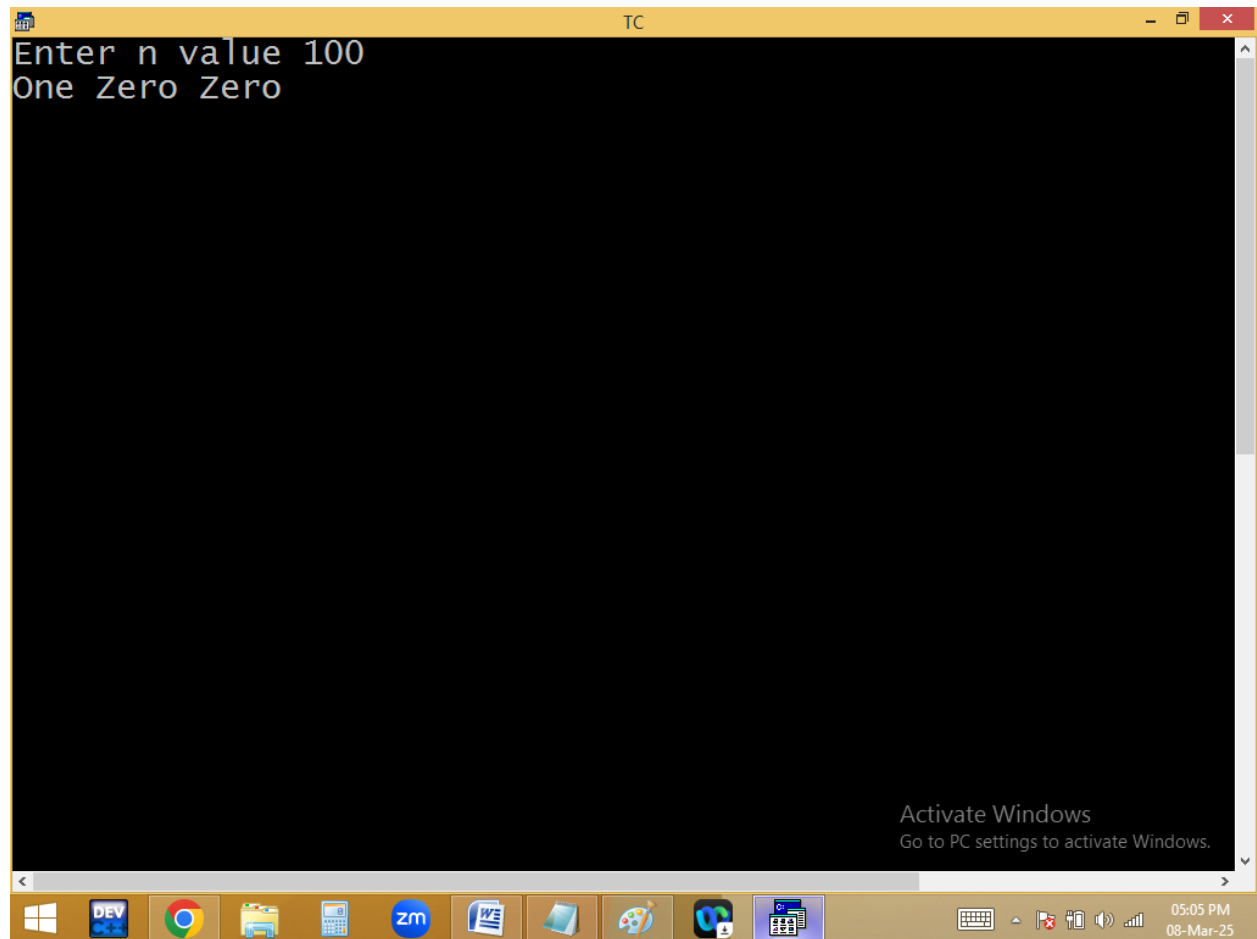
102 → One Zero Two

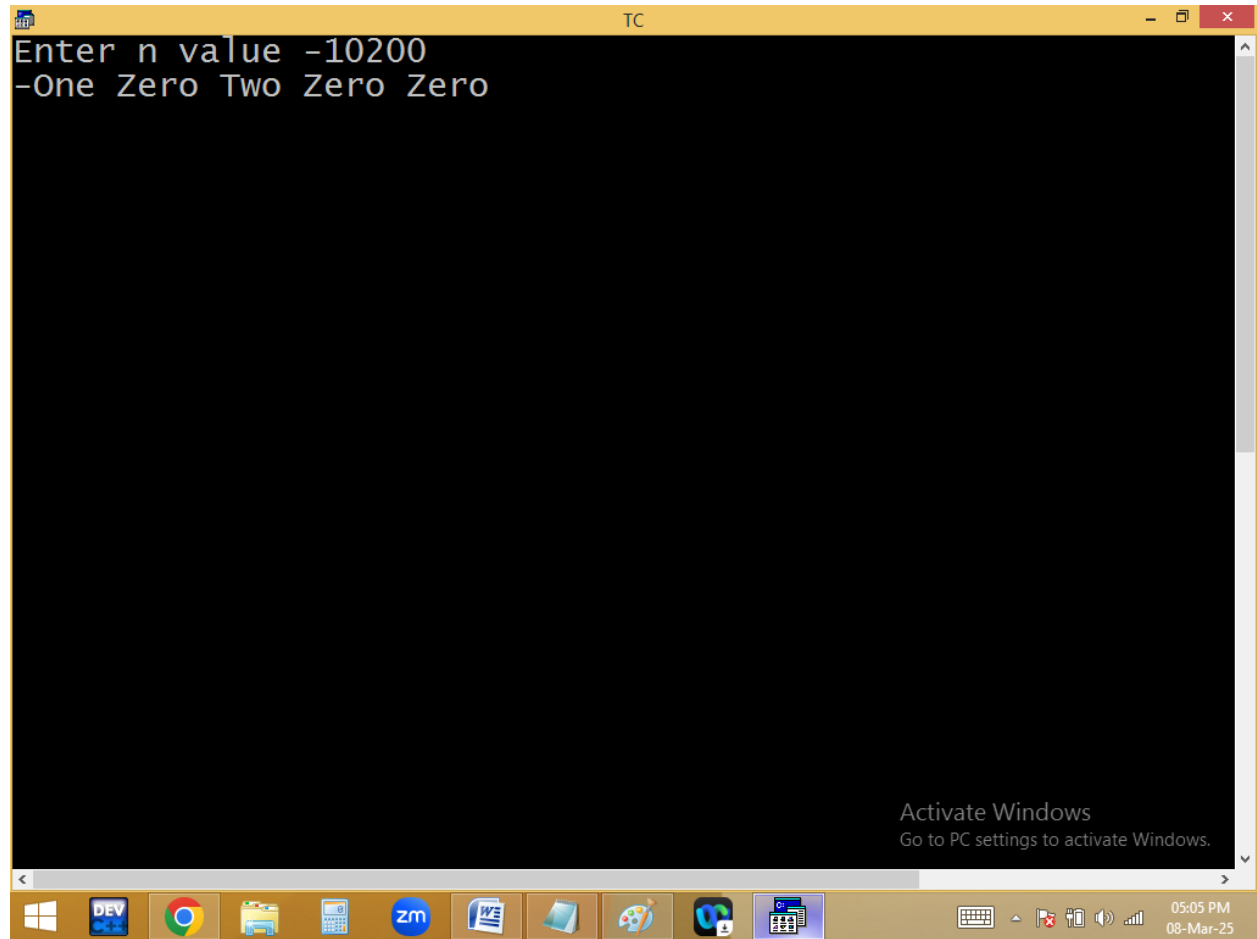


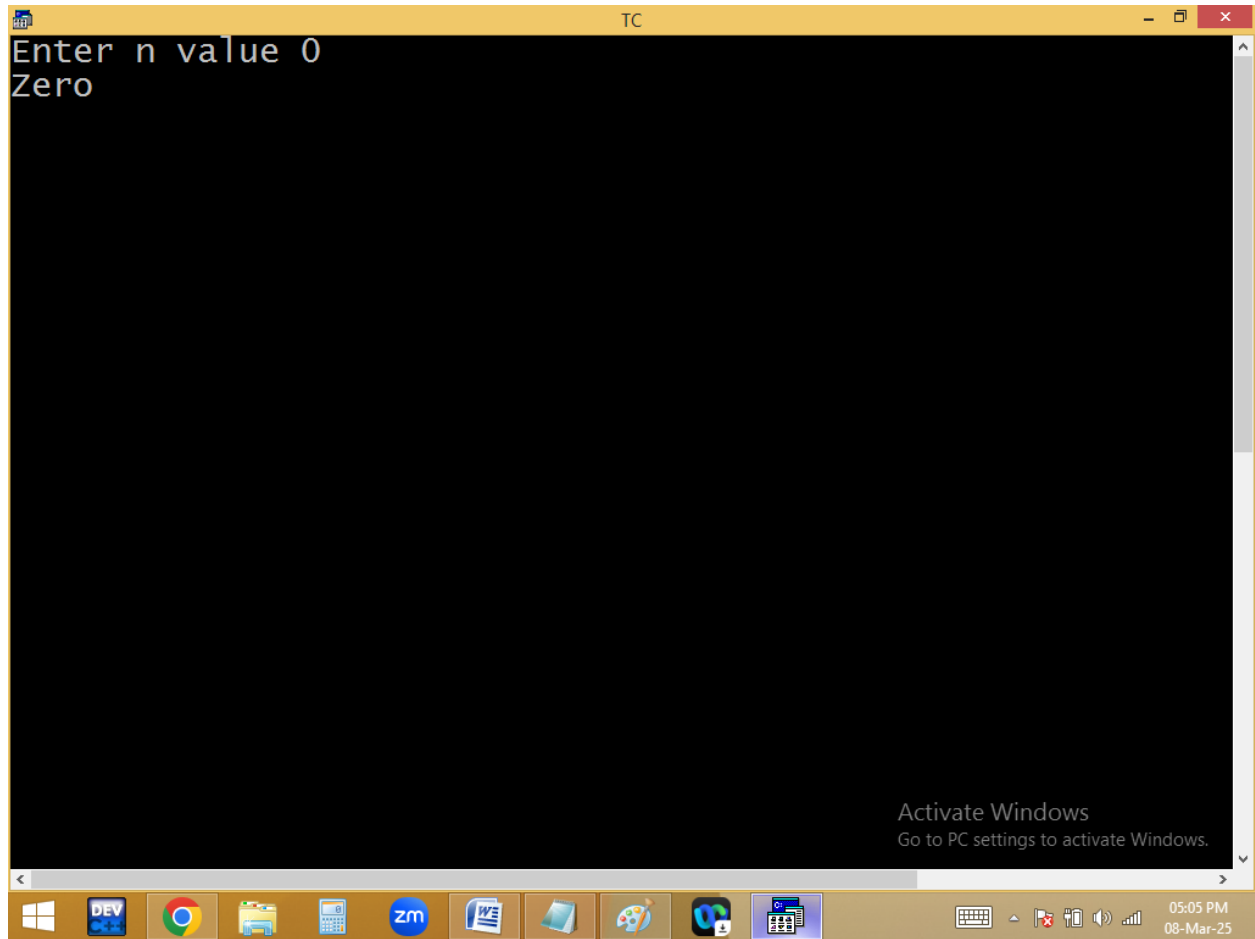
```
File Edit Run Compile Project Options Debug
Line 1 Col 53 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{ long m,n,rev=0;int r; clrscr();
printf("Enter n value "); scanf("%ld",&n);
if(n<0) printf("-",n=-n);m=n;
while(m){r=m%10;rev=rev*10+r;m=m/10;}//*rev no*/
do
{switch(rev%10)
{case 0: printf("Zero");break;
case 1: printf("One");break;
case 2: printf("Two");break;
case 3: printf("Three");break;
case 4: printf("Four");break;
case 5: printf("Five");break;
case 6: printf("Six");break;
case 7: printf("Seven");break;
case 8: printf("Eight");break;
case 9: printf("Nine");break;
} rev=rev/10; printf(" ");
} while(rev!=0);
while(n%10==0&&n!=0)printf("Zero ",n/=10); getch(); }
```

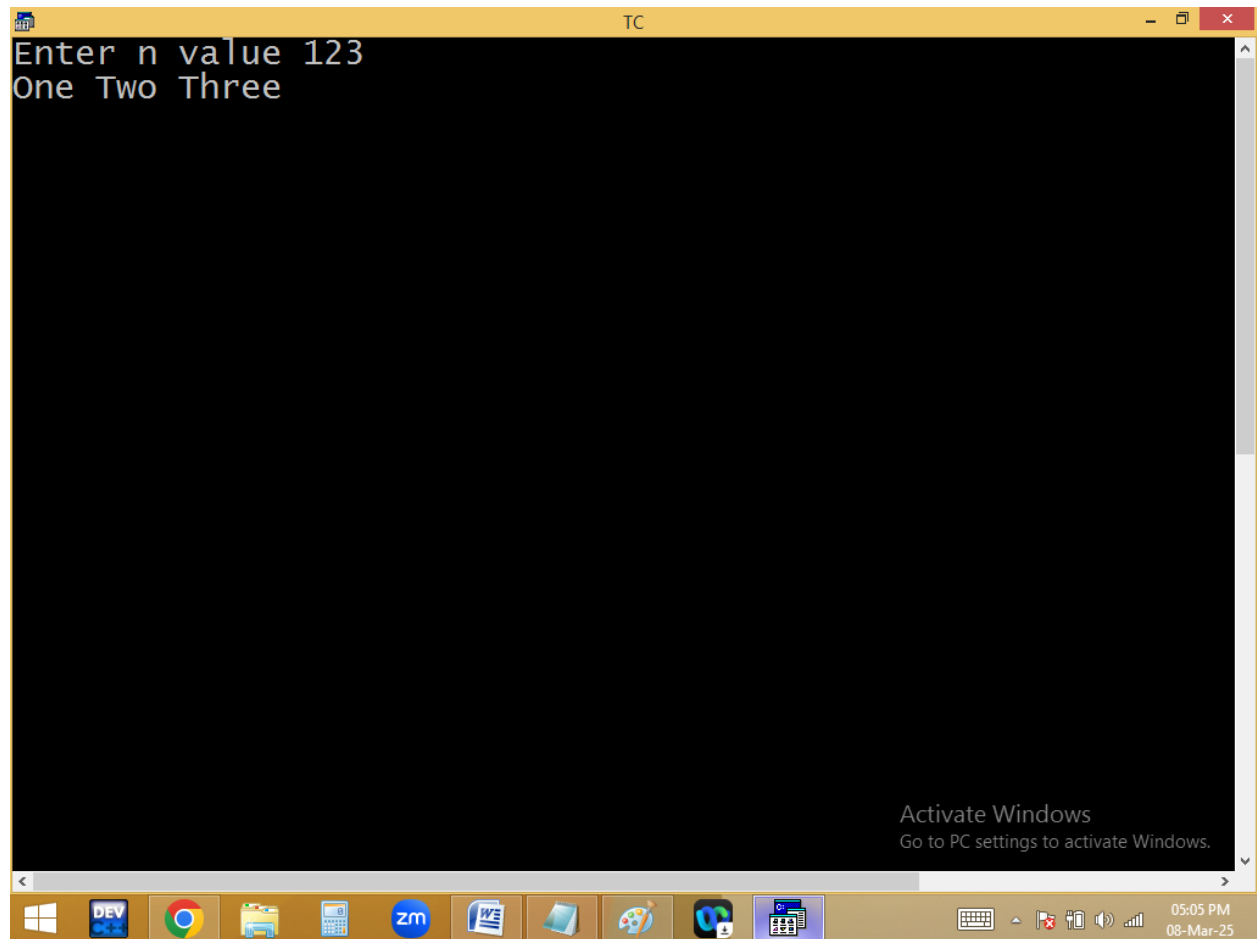
Activate Windows. Go to Settings to activate Windows.

05:05 PM 08-Mar-25









```
TC
Enter n value -10203000
-One Zero Two Zero Three Zero Zero Zero

Activate Windows
Go to PC settings to activate Windows.
```

```
do
{
switch(rev%10)
{
case 0: p(zero);b;
case 1: p(one);b;
case 2: p(two);b;
case 9: p(nine);
}
rev=rev/10;
p(" ");
}while(rev!=0);
```

n m
102 102

rev
 $201 \% 10 = 1$ one
 $20 \% 10 = 0$ zero
 $2 \% 10 = 2$ two
0
one Zero Zero

n
 $102 \% 10 = 0$
 $10 \% 10 = 0$
 $1 \% 10 = 1$

$\text{while}(n \% 10 == 0 \ \&\& \ n != 0) \ p(\text{"Zero"}, n / 10);$

Finding Armstrong no or not?

1 is a single digit no = $1^1 = 1$

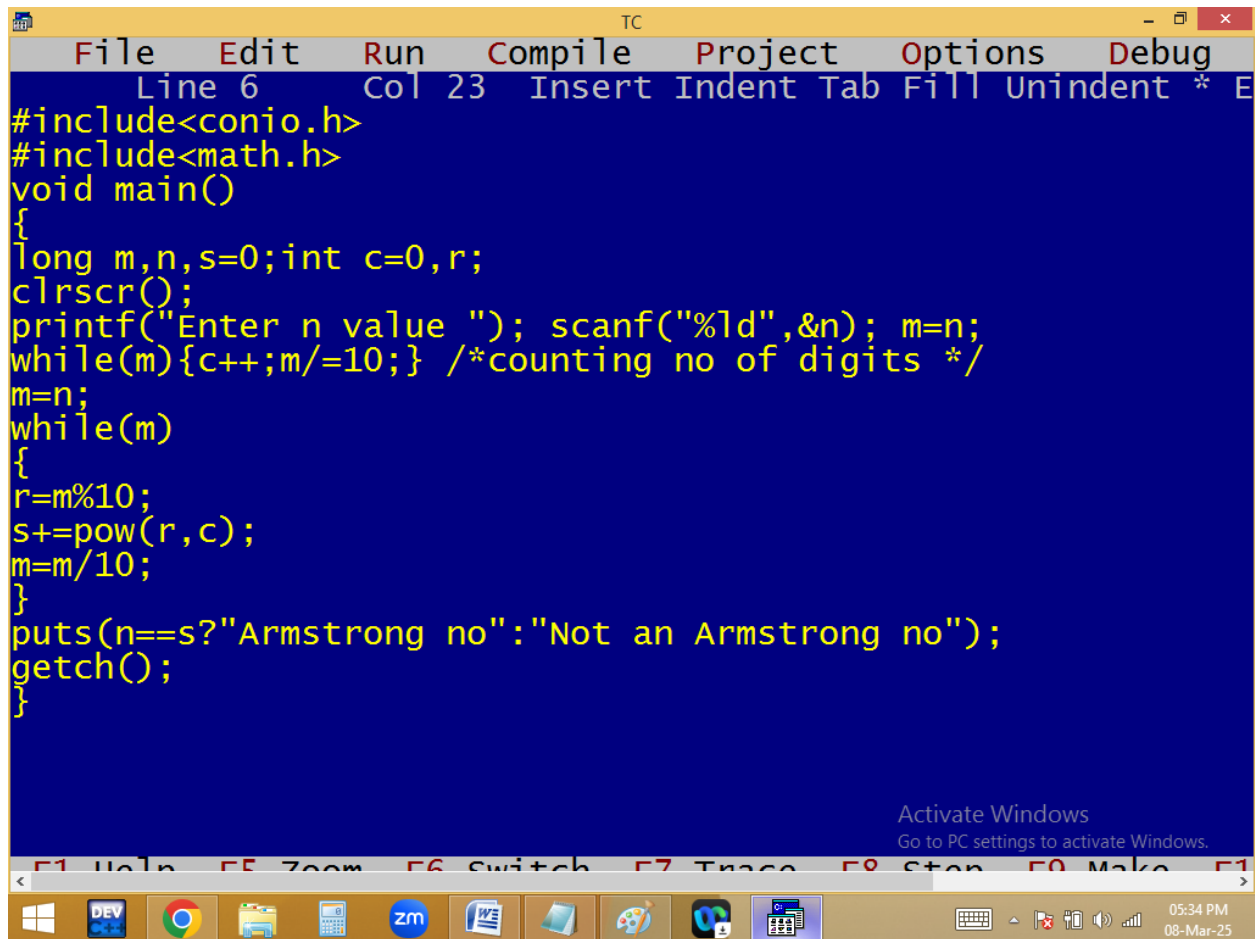
2 is a single digit no = $2^1 = 2$

9 is a single digit no = $9^1 = 9$

153 is a 3 digit no = $1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153$

370, 371, 407, 1634, 8208,...

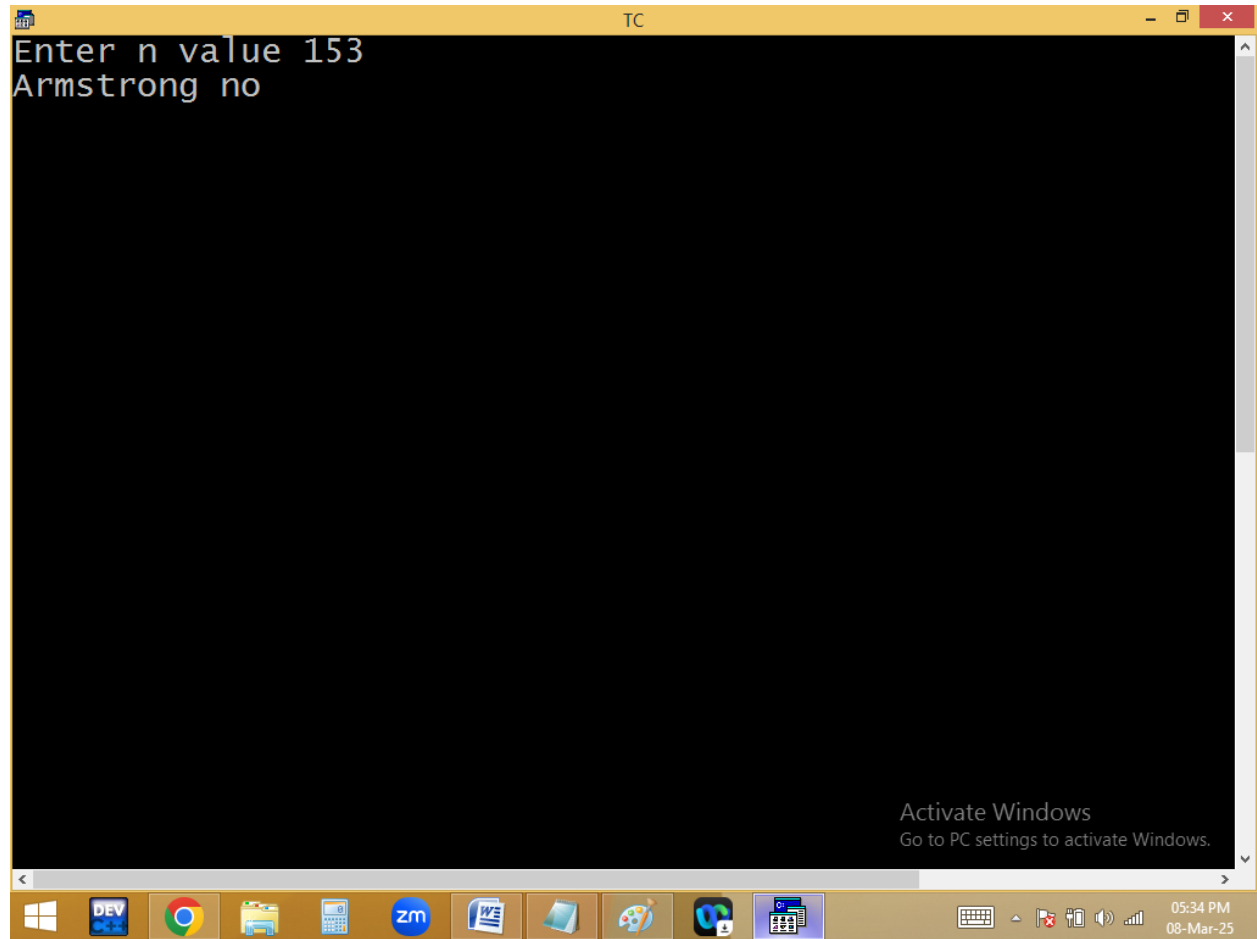
$1634 = 1^4 + 6^4 + 3^4 + 4^4 = 1634$

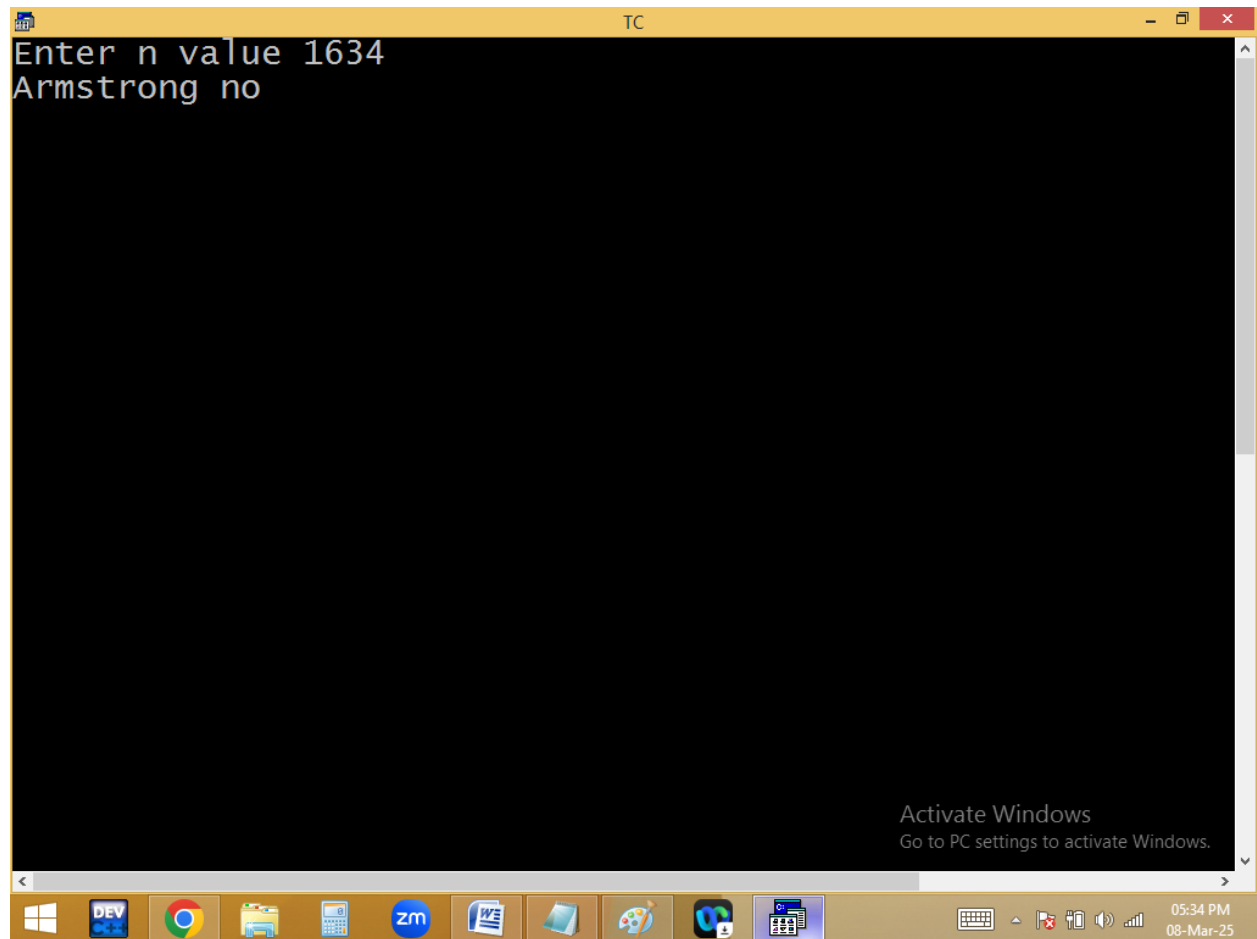


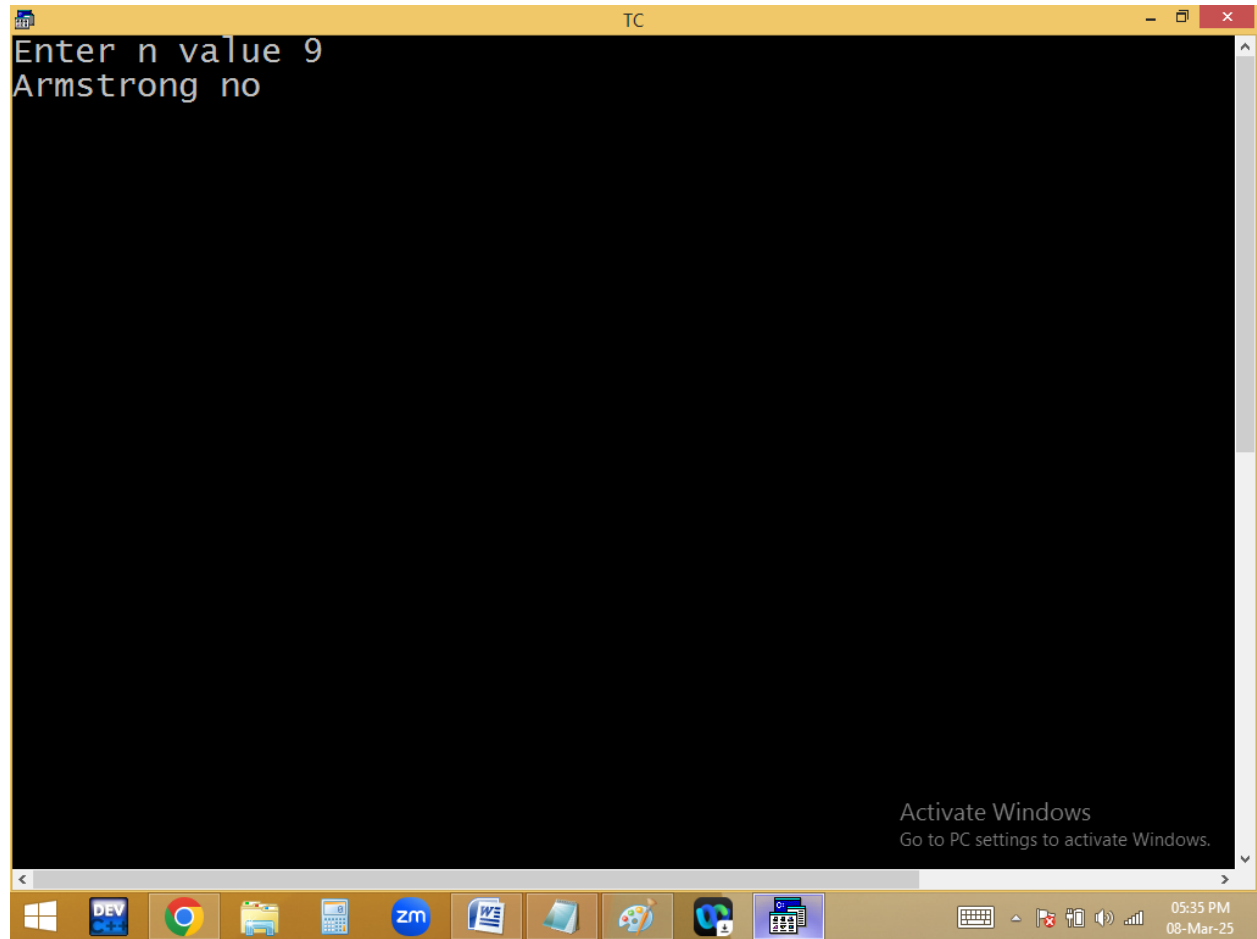
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 6", "Col 23", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", and "* E". The main editing area has a blue background and contains the following C code:

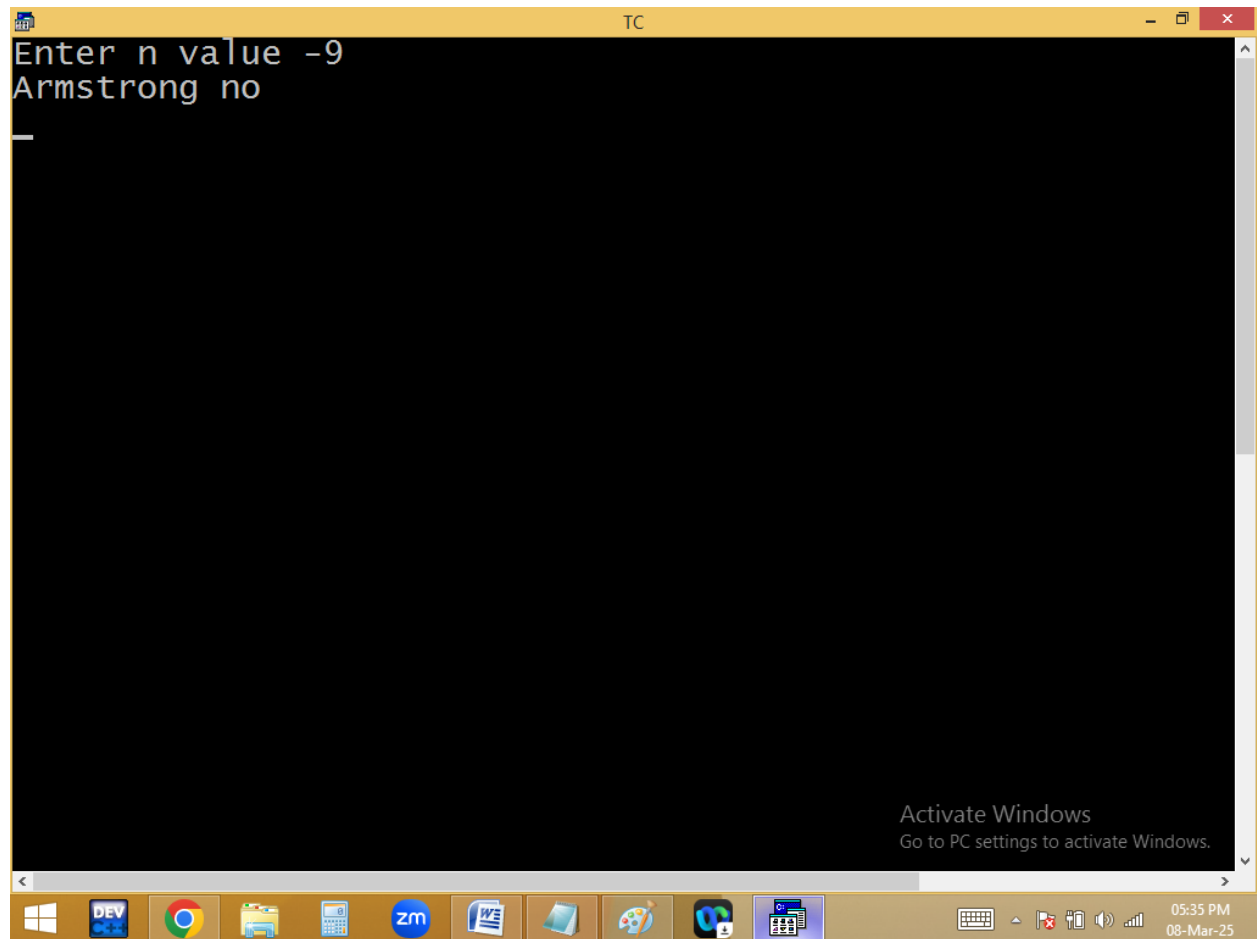
```
#include<conio.h>
#include<math.h>
void main()
{
long m,n,s=0;int c=0,r;
clrscr();
printf("Enter n value "); scanf("%ld",&n); m=n;
while(m){c++;m/=10;} /*counting no of digits */
m=n;
while(m)
{
r=m%10;
s+=pow(r,c);
m=m/10;
}
puts(n==s?"Armstrong no":"Not an Armstrong no");
getch();
}
```

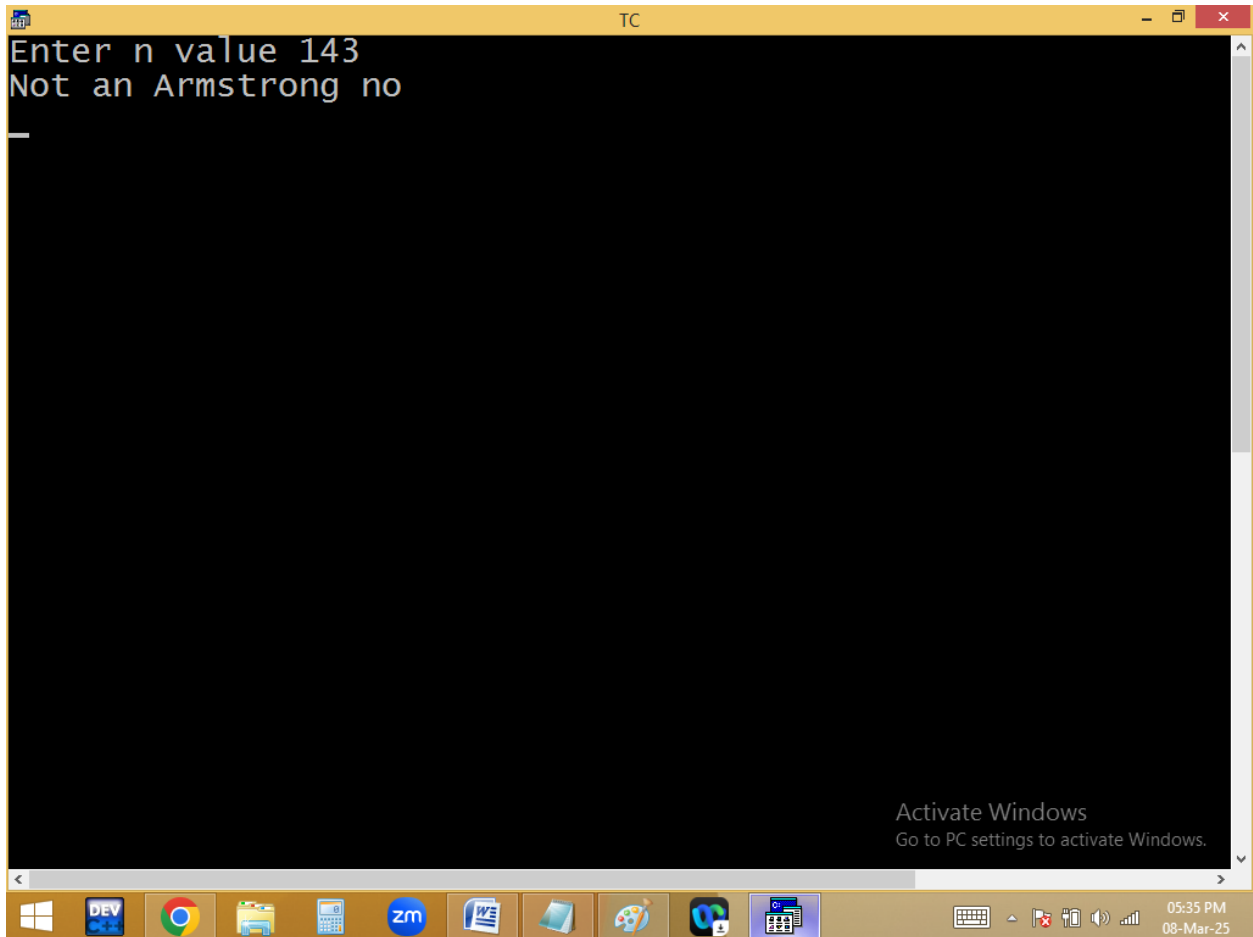
At the bottom of the IDE window, there is a text prompt: "Activate Windows. Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which includes icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, Word, and other applications. The system tray on the right shows the time as "05:34 PM" and the date as "08-Mar-25".











~~0 + 53~~
while(m){c++;m/=10;}

$$\frac{m}{153} = \frac{m}{153}$$

c = 0 1 2 3

m=n;
while(m!=0)
{
r=m%10;
s+=pow(r,c);
m/=10; ✓
}

$$\begin{array}{r} m \\ 153 \\ 153 \div 10 = 3 \\ 15 \div 10 = 5 \\ 1 \div 10 = 1 \\ 0 \end{array}$$

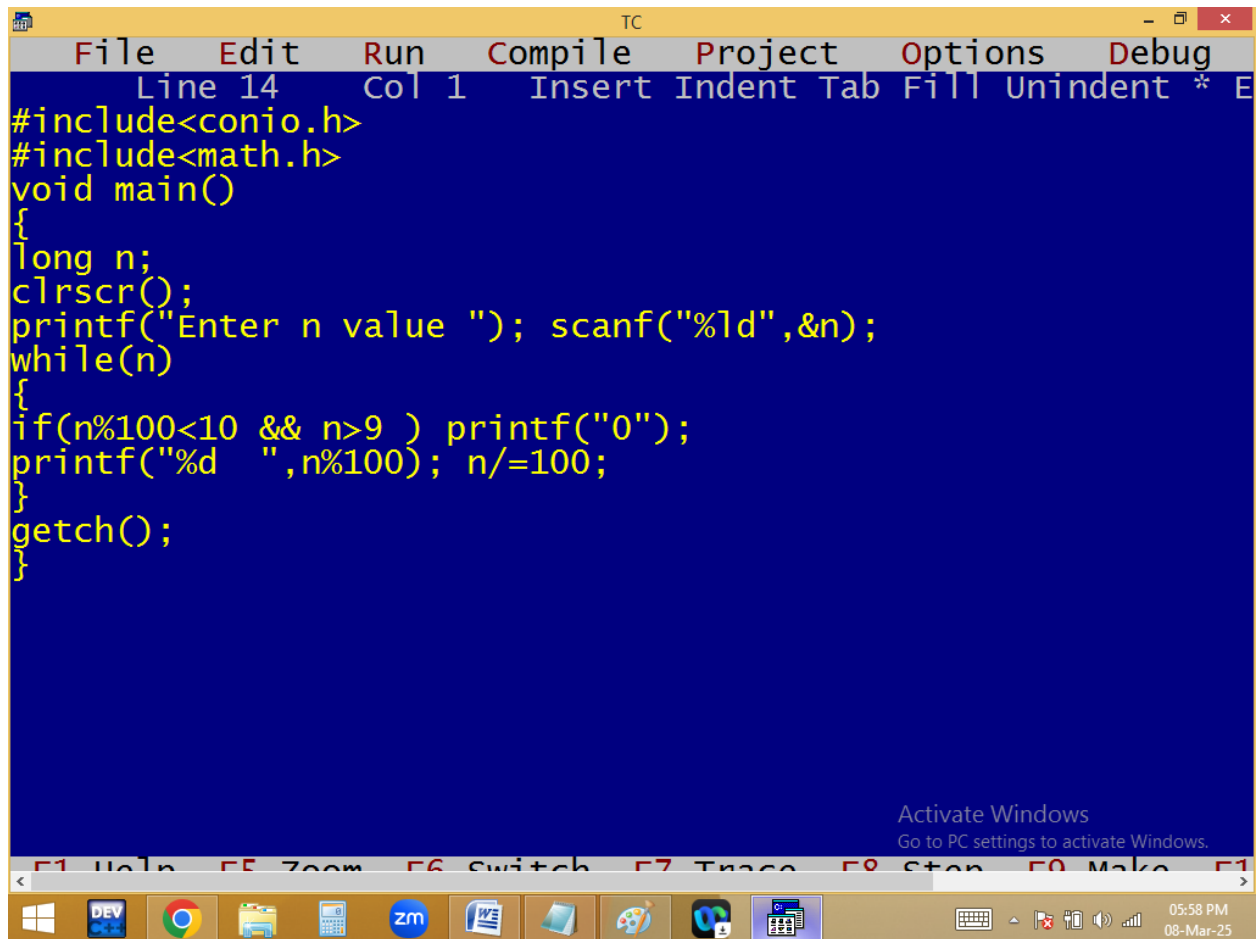
$$\begin{array}{r} s \\ 0 + 27 \\ + 125 \\ + 1 \\ \hline 153 \end{array}$$

puts(n==s?"Armstrong":"Not");

$$143 = 1 + 64 + 27 = 92$$

Print the given no as follows.

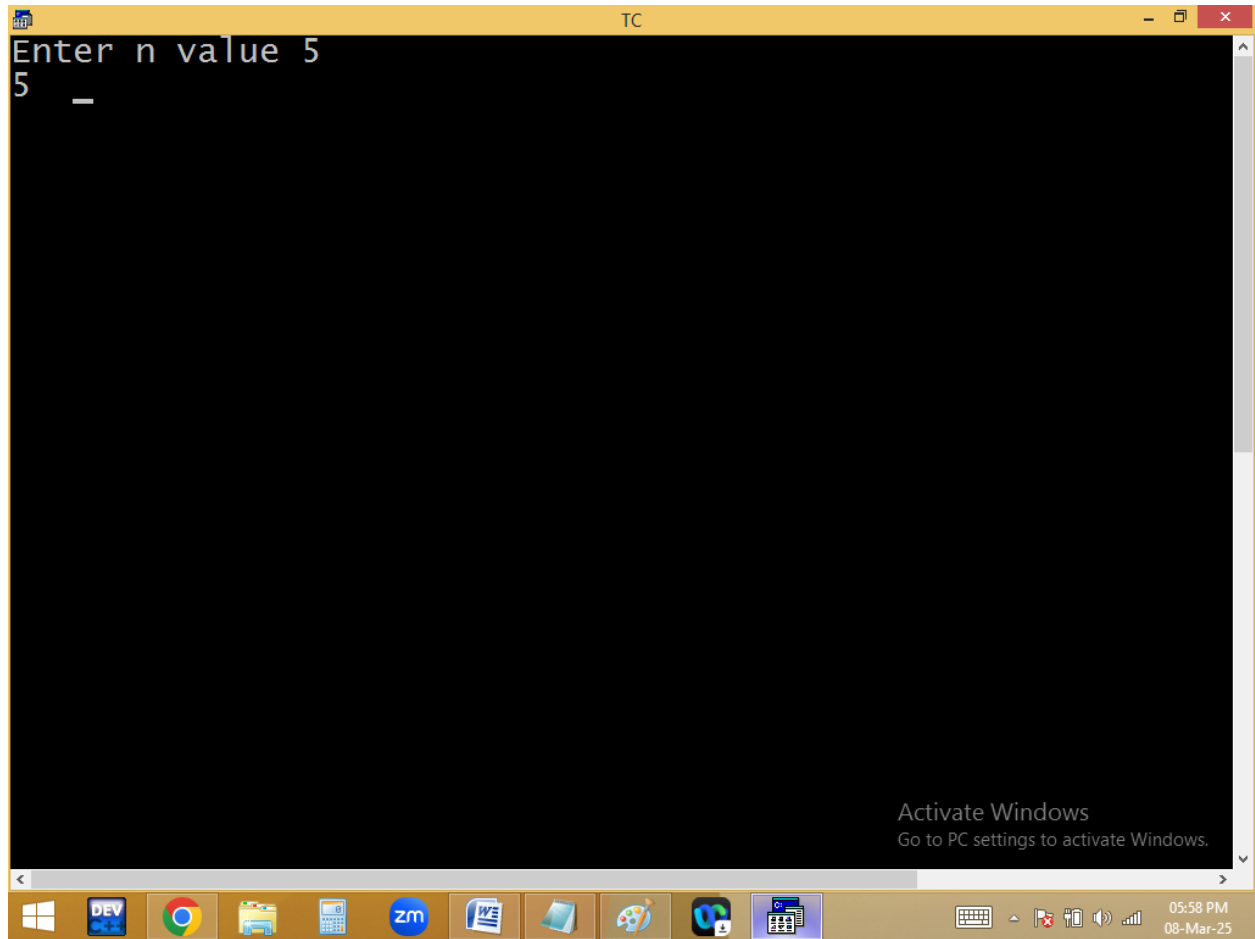
12345678 → 78 56 34 12



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 14 Col 1" and lists keyboard shortcuts: Insert, Indent, Tab, Fill, Unindent, *, and E. The main editing area has a blue background and contains the following C code:

```
#include<conio.h>
#include<math.h>
void main()
{
long n;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
while(n)
{
if(n%100<10 && n>9 ) printf("0");
printf("%d  ",n%100); n/=100;
}
getch();
}
```

At the bottom of the IDE window, there is a text prompt: "Activate Windows Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which includes icons for Windows, DEV, Chrome, File Explorer, Calculator, Zoom, Word, a folder, Paint, OneDrive, and a calendar. The system tray on the right shows the date and time: "05:58 PM 08-Mar-25".




```
TC
Enter n value 1000000
00 00 00 1
```

Activate Windows
Go to PC settings to activate Windows.

$$102 \% 100 = 2$$

`if(n%100<10 && n>9 ) p(0);`

2

$$123 \% 100 = 23$$

5

Home work:

1. Neon number – sqr of given no digits
sum is equal to given no

Eg: $9 \rightarrow 9*9=81 \rightarrow 8+1=9$

2. harshad number / niven number

18 is a harshad number because $1 + 8 = 9$,
and 18 is divisible by 9

1729 is a harshad number because $1 + 7 + 2 + 9 = 19$, and 1729 is divisible by 19

2025 is a harshad number because $2 + 0 + 2 + 5 = 9$, and 2025 is divisible by 9

Other names

$1234=1+2+3+4=10 \leftarrow$ not a harshad no

3. finding no of even/odd/0 digits in given
no

eg: 1002 $\rightarrow$ 1 even, 1 odd, 2 zero

4. finding the given digit position in given
no

eg: 107347 $\rightarrow$ 7 position is 3, 6

